Transportation Research Part C 147 (2023) 103994

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

Joint optimization of train scheduling and routing in a coupled multi-resolution space–time railway network ☆

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ARTICLE INFO

Keywords:
Train scheduling
Train routing
Joint optimization
Lagrangian relaxation
Branch-and-bound algorithm

ABSTRACT

Train scheduling and routing are two critical and fundamental problems in the field of railway operations, especially under the large demand scenarios or complicated operation schemes. These problems are often solved separately and sequentially from the perspectives of different decision stages. To further make these two decision-making processes consistent, this paper develops a joint optimization model in a multi-resolution space–time network, by coupling the conflict-free routing constraints into the train scheduling process. To solve this model, a feedback iterative framework consisting of the Lagrangian relaxation and the branch-and-bound algorithm is developed, in which the Lagrangian relaxation aims to generate the train schedule and the branch-and-bound strategy with the constraint updating procedure resolves the potential micro conflicts. Compared with some existing approaches in the literature, the proposed model and solution algorithm have the merit of no requirement of optimization solvers. To test the performance of the proposed approach, we apply the algorithm to different small-scale operation scenarios and a real-life case with the INFORMS RAS 2016 (Railway Applications Section) dataset.

1. Introduction

Rail transportation has advantages over other transportation modes in terms of speed, capacity, safety, and energy efficiency. With the rapid development of railway industry in recent years, the mileage of the railway network in China has reached 146,000 kilometers by 2020 (including 35,000 kilometers of high-speed railway and over 100,000 kilometers of common railway), which has become the backbone of medium and long-distance transportation. The railway transport involves a complex process, including the optimization and control of various factors such as passengers, locomotives and crews. To reduce the complexity of operation planning, a hierarchical strategy with various subproblems is often adopted in reality, shown in Fig. 1 (Lusby et al., 2011). In this procedure, the operation planning can be processed sequentially from different decision levels (i.e., strategic, tactical and operational levels), according to diverse network scales and planning horizons. Specifically, the service network design is firstly determined on the strategic level, based on the origin–destination (O–D) demand; on the tactical level, some off-line planing components, such as train scheduling, train routing, crew assignment, etc., should be accomplished in a sequential manner; finally, the involved trains need to be rescheduled on the real-time operational level, when the normal disruption occurs in the railway system.

☆ This article belongs to the Virtual Special Issue on IG005585: VSI:Advanced railway transpo.

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https://doi.org/10.1016/j.trc.2022.103994
Received 15 December 2021; Received in revised form 10 December 2022; Accepted 15 December 2022
Available online 31 December 2022
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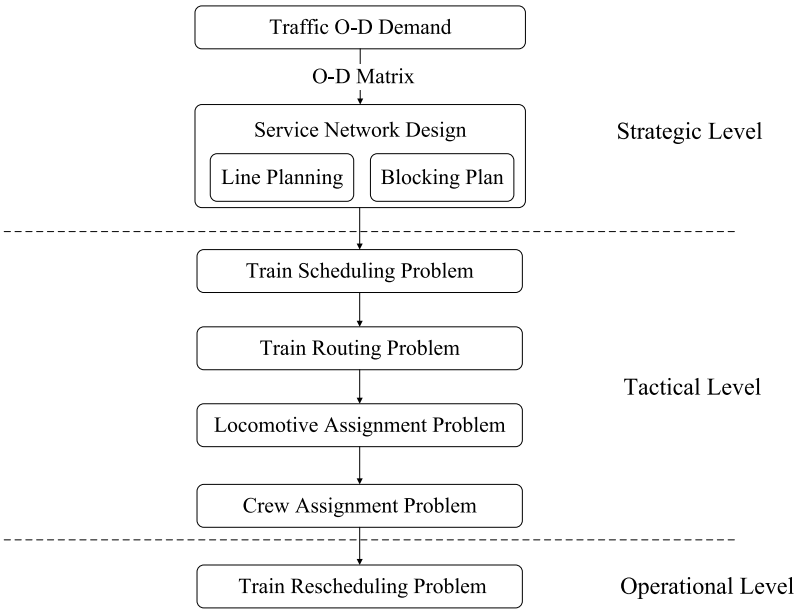


Fig. 1. Railway operation planning process.

Among different components of the operation scheme, the train schedule, which specifies a sketch of resource utilization plans and station departure–arrival times of trains according to the line planning scheme, plays a core role in the real-world operations. With a predetermined schedule, a train routing process then aims to find a set of conflict-free routes for individual trains in the turnout-based railway network (including the detailed track assignment, conflict resolution, etc.). The above decision processes are in general accomplished successively owing to their complex coupling relationships according to Lusby et al. (2011). For example, the dispatchers need to well allocate the railway track resources to individual trains under the information of the given train schedule so that the potential conflicts can be avoided at the junctions and platform tracks. However, it is hard in most cases to find the conflict-free routes manually for all trains in a busy and complex schedule. Thus, dispatchers have to feed back repeatedly to reschedule the macro train schedule to make these two issues consistent eventually.

Focusing on providing a system-optimal solution, this paper aims to investigate a joint optimization method for the train scheduling and routing problem to generate the consistent operation plan optimally. To this end, a multi-resolution space–time network is introduced to depict the joint optimization model. In detail, two space–time networks with different resolution levels are coupled together to characterize the spatial and temporal dynamics of train movements. The macro-resolution space–time network with a large time resolution (granularity) (e.g., 1 min) only contains the main framework of the railway system (e.g., stations and main tracks). The micro-resolution counterpart with a small time resolution (granularity) (e.g., 1 s) has more detailed topological structure (e.g., turnouts, track circuits, platform layout). In such a network, we intend to find the consistent scheduling and routing schemes simultaneously in consideration of their coupling relationships. It is worth mentioning that this "macro–micro" network based approach has also been adopted to handle a similar joint optimization problem in the literature, e.g., Schlechte et al. (2011), Bešinović et al. (2016), and Goverde et al. (2016). However, in comparison with these studies, the novelty of this paper is that we formulate our model in an arc-based space–time network combined with macro and micro features. Then, the Lagrangian relaxation (LR) and dynamic programming (DP) algorithms can then be embedded to solve this model effectively, without the aid of the optimization solver, which is different from the work (an integer linear programming (ILP) model with path variables) by Bešinović et al. (2016). With this method, it is much easier for us to formulate the problem of interest as a flow-based optimization model and generate more straightforward space–time trajectories for involved trains in macro and micro space–time networks. Besides, in the feedback process from micro to macro solutions, the constraints updating method (similar to Bešinović et al. (2016)) is further embedded into the branch-and-bound algorithm framework to resolve the potential micro conflicts. Specifically, the main contributions of this study are summarized as follows.

(1) We propose an integer linear programming (ILP) model for the joint optimization of train scheduling and routing problems. These two problems can be characterized by the space–time trajectory optimization process in a coupled multi-resolution space–time network. The objective is to minimize the total travel time in consideration of the constraints on incompatible arc sets and multi-resolution coupling relationships. Compared with the existing methods, we use the coupling constraints to integrate space–time resource decision variables across different levels. This proposed model can guarantee headway requirements in the macro-resolution network and conflict-free train routes in the micro-resolution network.

(2) A feedback iterative algorithm is proposed to solve the ILP model, in which the primal model is decomposed into two subproblems related to train scheduling and train routing. The algorithm starts with the train scheduling process. A DP-based LR

algorithm is designed to find a train schedule in the macro-resolution network, and this schedule is used to find paths for different trains in the micro-resolution network. If conflicts occur on micro paths, a branch-and-bound (BB) framework with constraint updating (CU) is used to estimate the optimal candidate set of headways between conflict train pairs, which is designed to reschedule the previous scheduling scheme. This feedback process keeps updating until feasible scheduling and routing schemes are produced simultaneously.

(3) To validate the proposed methods, a series of numerical experiments are conducted on a small-scale railway line with the sampled data and a real-world case with the INFORMS RAS 2016 dataset (Meng, 2016). The experimental results show that the proposed algorithm has the advantage of balancing computational complexity and solution accuracy of the joint optimization problem. Moreover, some managerial insights to improve the railway throughput capacity are also proposed by performance analyses with respect to critical parameters, such as, paying more attention to homogeneous factors in line planning schemes, coordinating the station layouts, utilizing the track resources flexibly, etc.

The remainder of this paper is organized as follows. Section 2 reviews a series of relevant literature on train scheduling and routing problems. Section 3 gives a detailed problem statement and representation of a coupled multi-resolution space–time network. A joint optimization model with multi-resolution coupling constraints is proposed in Section 4. Then, the detailed solution algorithms to solve the model are introduced in Section 5 and numerical experiments are conducted to verify effectiveness of the model in Section 6. Finally, some conclusions and further research directions are given in Section 7.

2. Literature review

Train scheduling and train routing are two critical decision components in developing a railway operation scheme, which have attracted tremendous attention from both academia and industry in recent decades (see Caprara et al. (2002) and Lusby et al. (2011) for detailed reviews). In the following discussion, we focus on introducing the state of the art in terms of three aspects, including train scheduling, train routing, and joint optimization of these two problems.

(1) Train scheduling problem. The train scheduling problem is to determine train arrival and departure times at each station in consideration of safety operations. In the literature, two effective modeling frameworks are proposed to characterize this problem, i.e., the job-shop scheduling model with big-M constraints and multi-commodity flow model with space–time network representation.

In the job-shop scheduling framework, the involved trains and tracks are viewed as a series of jobs and machines respectively. To model this problem as a mixed-integer linear programming model (MILP), the big-M method is often employed to handle the "if-then" constraints involved in some specific operation schemes, such as the stopping plan with passenger demands (Yang et al., 2016), real-time train rescheduling (Zhan et al., 2015; Zhang et al., 2020b), train maintenance plans (Zhang et al., 2019a) and dynamic passenger flow control strategies (Yuan et al., 2022). Besides, some variations of models based on the job shop scheduling framework are also proposed for this type of problem. For example, Zhu and Goverde (2019) discussed the application of the activity-event models in the railway rescheduling problem during disruptions, based on discrete activity events that a train departs from/arrives at stations. Besides, Boroun et al. (2020) proposed blocking permutation flexible flow shop scheduling model for the train scheduling and stop planning problem with the consideration of station capacity and time-dependent dwell time constraints.

In the space–time network modeling approach, some complicated constraints (e.g., time windows, pickup and delivery, and capacity constraints) can be embedded into the discrete spatial and temporal network. Then, the model can be decomposed into a sequence of shortest path problems as an approximation (Yue et al., 2016; Luan et al., 2017; Meng et al., 2020; Zhang et al., 2020a). Compared with the job-shop scheduling model, this method could generate the specific train trajectories precisely with the physical and time dimensions. Based on the space–time network, Brännlund et al. (1998) firstly proposed a segment-based LR method and improved the updating efficiency of the Lagrangian multipliers by using the sub-gradient method. Caprara et al. (2002) proved that the single-track train scheduling problem is an NP-hard problem, and the proposed departure–arrival node based model further improved the solution effectiveness of the arc-based model in the previous research by Brännlund et al. (1998). In recent years, many researchers further investigated the train scheduling related problems within the extended space–time network frameworks, such as the speed profile optimization with a space–time-speed network (Zhou et al., 2017), the train rescheduling problem with passenger routing (Zhan et al., 2021) and the joint optimization of train timetables, speed levels and stop plans based on the multi-layer space–time network for better describing element characterizations in the actual operations (Chen et al., 2022).

In addition, another challenge for the train scheduling problem is to design highly efficient solution algorithms. Many studies in the literature focus on exact algorithms for some small/medium-scale cases, which are mainly designed on the basis of column generation (CG) (Cacchiani et al., 2008; Tian and Niu, 2020), Dantzig–Wolfe (DW) decomposition (Min et al., 2011), Benders decomposition (BD) algorithms (Lamorgese and Mannino, 2015), etc. Since it is difficult to design exact algorithms for the large-scale train scheduling problem, some decomposition-based approximate algorithms have been proposed, where the LR algorithm is a commonly used heuristic (Brännlund et al., 1998; Luan et al., 2017; Zhou et al., 2017; Meng et al., 2020; Zhang et al., 2020a). Along this line, Brännlund et al. (1998) explored a train scheduling problem on the single-track line, and designed a LR with flow balance constraints and capacity constraints. In addition, some customized techniques can also be incorporated to improve the performance of LR in the process of solving the scheduling problem, e.g., dynamic programming (DP) (Luan et al., 2017) and dynamic constraint generation strategies (Zhang et al., 2020a).

(2) Train routing problem. Train routing can be regarded as a problem of allocating conflict-free resources for all trains in a micro-resolution space–time network. In some existing literatures, e.g., Zhang et al. (2019b, 2020c), the micro train scheduling or platform assignment problem can also be regarded as a special case of train routing problems, which determines the feasible

allocation sequence of station junctions and departure–arrival routing schemes for involved trains. Recently, the train routing problem is often formulated by various representations, such as the conflict graph, alternative graph, and space–time network. For instance, D'Ariano et al. (2007) presented a branch-and-bound approach to minimize the length of the longest path in an alternative graph, which was tested and compared with several heuristic methods. Furthermore, based on a pattern conflict graph, Caprara et al. (2011) developed an integer programming (IP) model to solve the micro train platform scheduling problem. In such a graph, nodes represent route patterns, and two involved nodes are incompatible if a connected link exists between these nodes. In addition, Caimi et al. (2011) used a conflict tree method to enumerate all possible micro train paths in the train routing problem. Based on the binary search tree (also known as the reconstructed resource conflict tree), the non-conflicting train paths were determined by solving the largest non-conflicting subset problem. Using a space–time network representation, Meng and Zhou (2014) adopted a solution algorithm on the basis of cumulative flow variables to minimize the deviation between the rescheduled plan and the original plan in micro train routing selection with maintenance time windows. In the maintenance track scheduling problem, Zhang et al. (2019b) used the big-M-based MILP to model the micro train routing problem. An iterative optimization algorithm for alternately fixing train paths and maintenance task time windows was designed to achieve an exact solution efficiently.

(3) Joint optimization of train scheduling and routing problems. In general, the considered railway network for train scheduling is mainly composed of stations and segments, while the railway network for routing, which has a more complicated micro topological structure, consists of detailed turn-outs nodes and tracks. In this sense, the space and time granularity in the network of train routing is more delicate than that of train scheduling. In operation planning, since any conflict between train routes should be avoided in the train timetable, many studies begin to consider the influence of microscopic conflicts of routes when investigating the train scheduling problem. Accordingly, two approaches, including "macro–micro" and mesoscopic networks based methods, are developed to solve the joint optimization of train scheduling and routing by representing the railway network with different resolutions.

In recent years, the "macro–micro" networks based approach has drawn much attention in the literature (Zhou et al., 2022). In Schlechte et al. (2011), the results of micro train routing simulation can be converted into the macro counterpart through the network aggregation. After the macro scheduling, an optimal micro train routing scheme can be simulated through OPENTRACK. Zhu et al. (2014) studied a scheduled service network design problem in freight railway at the macroscopic level, which includes an integrated plan of service selection and scheduling, car classification and blocking at the microscopic level. As for the train dispatching problem in Lamorgese and Mannino (2015), Lamorgese et al. (2016), an exact Benders-like decomposition approach was developed with the macro and micro representations of railway networks. The similar approaches were also extended to solve the cyclic and non-cyclic train timetabling problem (Lamorgese et al., 2017). Bešinović et al. (2016) proposed a hierarchical framework for robust train scheduling, which optimized the robust and path-based schedule at the macro level, and generated a feasible and stable timetable at the micro level. Goverde et al. (2016) proposed a scheduling operation framework with a benefit evaluation model. A multi-layer framework (including macro level, micro level, and rescheduling level) was used to integrate relevant optimization factors for a railway system to maximize the robust feasibility of a global optimization.

Joint optimization for train scheduling and routing problems can also be formulated at the mesoscopic level, in which the railway network consists of detailed station structure and track segments. The station structure includes not only the platform tracks but also the tracks in the station bottleneck areas, and a track segment between two stations is viewed as a unique segment if it contains no turnouts. In a mesoscopic resolution space–time network, Zhang et al. (2020c) proposed a simultaneous optimization approach for re-optimizing the train and platform schedules to minimize the train cancellation cost and total travel time. In experimental results, it is demonstrated that an alternating direction method of multipliers (ADMM)based approach is able to provide a tighter upper bound than the LR approach. In addition, Zhang et al. (2022) simultaneously optimized train timetabling, platforming, and network maintenance scheduling for a railway network on a mesoscopic level.

Next, to highlight the innovations of this study, Table 1 compares some recent works regarding the joint optimization problem and the "macro–micro" approach, in terms of space and time resolutions, solution algorithm and route feasibility. The key differences between this research and existing ones can also be summarized as follows.

(a) *Modeling framework*: compared with the MILP models of the joint optimization problem by Schlechte et al. (2011), Bešinović et al. (2016) and Goverde et al. (2016), this paper formulates the specific train trajectories in the space–time network with flow balance constraints, which could be easily decoupled by decomposition algorithms and solved with the highly efficient shortest path algorithms (e.g., labeling setting/correcting algorithm, dynamic programming). (b) *Network resolution*: our study considers micro routes based on the turn-outs and second-oriented time horizon, which can guarantee conflict-free micro routes at station junctions. In comparison, some previous studies (Goverde et al., 2016; Zhang et al., 2020c) might not necessarily generate fully feasible micro train routes due to the settings of time resolutions. (c) *Solution approach*: the algorithm is designed in this paper to generate space–time trajectories of trains by using LR and DP, which is able to obtain the near-optimal solution without using any MILP solvers in comparison with some solver-based solution methods by Schlechte et al. (2011), Bešinović et al. (2016) and Goverde et al. (2016). In addition, some customized railway operation schemes can be flexibly embedded into the adopted space–time network without imposing additional constraints, e.g., time windows constraints (Mahmoudi and Zhou, 2016; Zhang et al., 2019c; Yao et al., 2019), speed profile constraints (Zhou et al., 2017), etc.

3. Problem statement and network modeling

3.1. Problem statement

The multi-resolution space–time network for train scheduling and routing is composed of two frameworks that represent the same railway network but with both macro and micro resolutions of the details. Denote the macro railway network as $G^M = (N^M, A^M)$,

Table 1
Comparison of some closely related studies in the literature.

Studies	Model	Physical resolution	Time resolution	Feasible micro-routes?	Solution method
Schlechte et al. (2011)	MILP	(Macro) Main timetable node (Micro) Infrastructure node	(Macro) 1 min (Micro) 6 s	Y	CPLEX
Bešinović et al. (2016)	MILP	(Macro) Main timetable node (Micro) Infrastructure node	(Macro) 1 min (Micro) N/A	Y	Heuristic; Constraint Updating
Goverde et al. (2016)	MILP	(Macro) Main timetable node (Meso) Train trajectory (Micro)	(Macro) 1 min (Meso) N/A (Micro)	N	Railway-ML System
Lamorgese et al. (2017)	MILP	Infrastructure node (Macro) Main timetable node (Micro) Infrastructure node (Meso)	1 min (Macro) 1 min (Micro) N/A (Meso)	Y	Benders-like decomposition
Zhang et al. (2020c)	STN	Platform node (Macro)	1 min (Macro)	N	ADMM
This paper	STN	Main timetable node (Micro) Infrastructure node	1 min (Micro) 1 s	Y	LR+BB; Constraint Updating

Symbol descriptions for Table 1:
Time resolution: only for the largest case study (N/A: Not available for specific resolution information).
Model: Space–time network (STN)/Mixed-integer linear programming (MILP).
Resolution/granularity: Macroscope (Macro)/Mesoscope (Meso)/Microscope (Micro).
Algorithm: Lagrangian relaxation (LR)/Alternating direction method of multipliers (ADMM)/CPLEX solver (CLPEX)/ Dynamic programming (DP)/Branch-and-bound (BB).

where N^M and A^M are the sets of stations and segments, respectively; I and J are used to denote the indexes of stations in the macro railway network (i.e., $I, J \in N^M$); (I, J) is the index of the segment from station I to station J (i.e., $(I, J) \in A^M$), as shown in Fig. 2(a). Besides, the macro tracks in upstream and downstream directions are defined as unidirectional links. Based on this network, a train schedule is first generated to guide the movements of involved trains, as illustrated in Fig. 2(b). At the operation level, a micro railway network should be taken into consideration to avoid potential conflicts between different trains at stations. Next, we denote the micro railway network as $G^m = (N^m, A^m)$, where micro nodes $i \in N^m$ are either turn-outs or signals, and micro links $(i, j) \in A^m$ correspond to track circuits between two adjacent micro nodes i and j. In operations, trains are allowed to dwell at a station on a departure–arrival track (e.g., tracks 1, 4, and 5 in Fig. 2(c)) and pass through a station via the main track (e.g., tracks II, III in Fig. 2(c)). Then, a micro track can be either unidirectional or bidirectional according to the specific scheme.

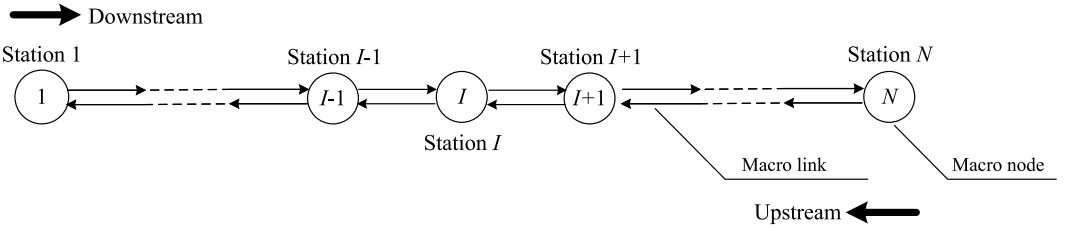
Next, we consider an example with the specific operation scheme in Fig. 2 to illustrate the inconsistency between macro and micro railway networks. In Fig. 2(c), station I, an intermediate station on this railway corridor, is bounded by a set of micro nodes (i.e., G_0, G_1, G_2, and G_3). A feasible train schedule at the macro level is illustrated in Fig. 2(b), which fully satisfies the headway and running time constraints. However, it is not feasible at time period [7 min, 8 min] in the micro network, as shown in Fig. 2(c), i.e., trains 1 and 2 will conflict at station junction A.

Based on the fixed and pre-given macroscopic train schedule (Fig. 2(b)), train 1 needs to arrive at station I at timestamp 7 min, while train 2 needs to depart from station I at timestamp 8 min. Therefore, the blocking time of train 1 occupying junction A lasts in time interval [6 min, 8 min] (Green shadow box in Fig. 2(b)), including 1 min buffer time before and after arrival time (i.e., 7 min). Similarly, the blocking time of train 2 lasts in time interval [7 min, 9 min] (Blue shadow box in Fig. 2(b)). These two blocking periods overlap in time interval [7 min, 8 min], resulting in a possible conflict under the station-based blocking condition. Thus, without considering the track allocation at stations, a macro train schedule might lead to the aforementioned conflicts and thus need to be rescheduled in order to generate feasible dispatching operations.

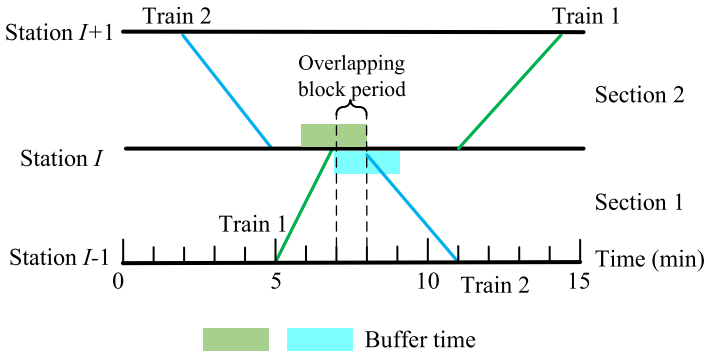
In reality, there are usually two possible reasons for the occurrence of conflicts mentioned above, which are (1) a high-density train schedule (e.g., shorter headway on segments) might lead to the excessive burden on the limited station capacity, and (2) complicated operation schemes in railway stations might also cause the overlapping of train space–time routes in junctions. Next, we give some illustrations below to further clarify these two possible reasons.

- **Case I: High-density train schedule and limited station capacity**

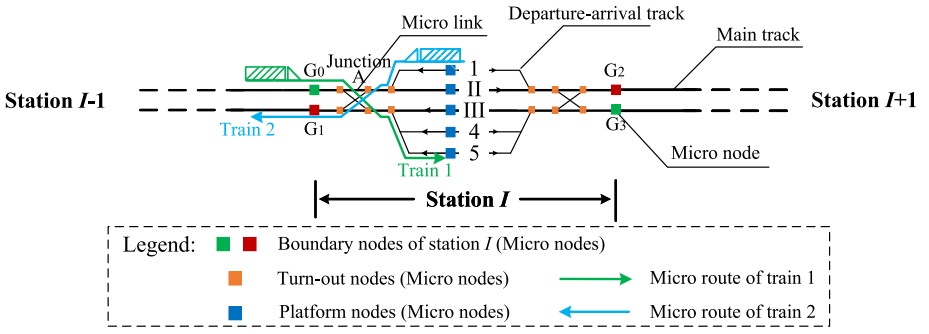
In practice, a train schedule with a 3-minute headway is widely used in high-speed railway systems to improve transportation efficiency. However, some trains cannot follow such a headway because of the limited track capacities and complex junction turn-outs at stations. In this case, the following trains have to wait in queue outside the station until the junction is cleared. For example, a schedule is illustrated in Fig. 3(a), where train 2 is required to follow train 1 with a 3-minute headway on the segment. If such a



(a) Macro railway physical network



(b) Macro railway operation network (Train schedule)



(c) Micro railway physical network of station I

Fig. 2. Mapping between macro and micro railway networks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

pre-given train schedule is executed, only micro track 1 can be used for downstream trains and the minimum safe operation time is 4 min. Then, an overlapping blocking region will emerge in Fig. 3(b), which results in the conflict between trains 1 and 2, i.e., the macro and micro schedules are practically infeasible due to the occurrence of this conflict region. In this scenario, train 2 needs to wait one more minute before it can enter track 1 to obtain a conflict-free train schedule.

• **Case II: Complicated operation modes in stations**

Compared with passenger rail, freight rail has a more flexible track allocation strategy. Some classification yards or major stations often allow trains to travel upstream (or downstream) and stop on downstream (or upstream) tracks in case of heavy traffic load, which corresponds to a flexible track usage strategy. In many circumstances, some trains need to undergo specific operations at these stations. For example, one train is scheduled to change the locomotive, but only tracks 4 and 5 can provide such a service, as

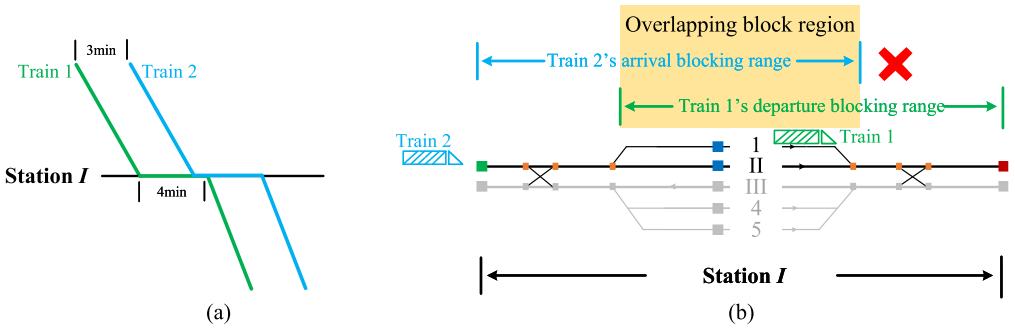
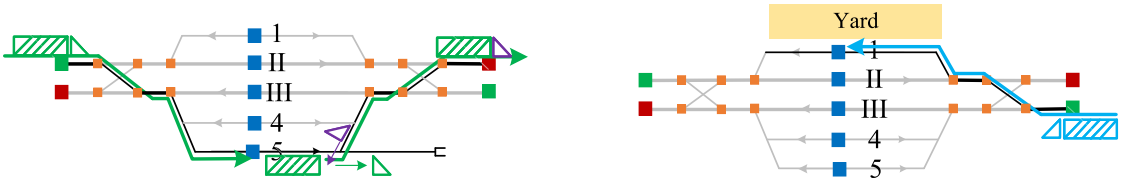


Fig. 3. Conflict of trains traveling in the same direction.



(a) Train routes for changing locomotives at track 5 (b) Train routes for loading/unloading freights at track 1

Fig. 4. Train routes with different station operation patterns.

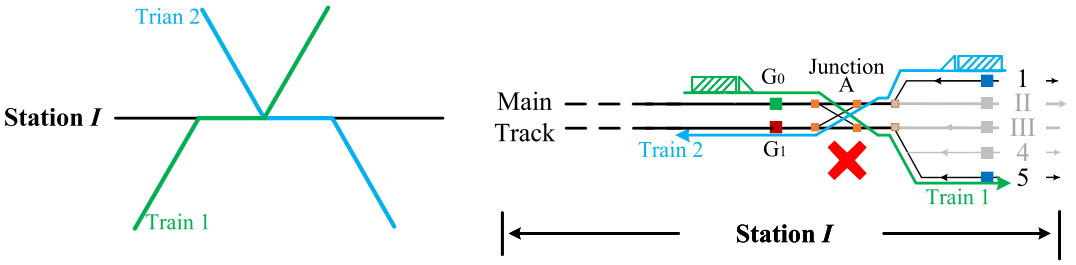


Fig. 5. Conflict of trains traveling in the opposite direction.

shown in Fig. 4(a). So, this train has to choose arrival and departure routes that cross the main track. Another example in Fig. 4(b) is that, a train needs to load or unload freights at the station while the freight yard is laid to one side of the station and only track 1 is available for this train. Then, this train needs to cross the main track at the junction. Nevertheless, since the train scheduling process does not consider the flexible track utilization and often plans the schedules separately in individual directions, it is difficult to locate such situations precisely in the macro train schedules in reality. Thus, if such a strategy is put into operation, as shown in Fig. 5, a potential conflict between train 1 arriving at track 5 and train 2 departing from track 1 will happen at junction A under this schedule. Also, if these trains with diverse route patterns are not scheduled by appropriate rules, some losses might occur with respect to the railway operation efficiency.

Two methods can be employed to resolve these issues and obtain fully conflict-free train space–time routes. (1) We can solve the train routing scheme directly in the corresponding micro railway network, which is only effective for some small-scale instances. (2) We can guarantee the right of way for individual trains at each junction by enlarging the passing headway; however, this approach will necessarily lead to the waste of time resources, especially in some complicated turn-out throats. With these concerns, this paper aims to use a coupled multi-resolution space–time network to balance the problem size and solution accuracy.

3.2. Assumptions

Without loss of generality, this section will introduce the following assumptions with respect to the attributes of railway networks and trains in the proposed mathematical model.

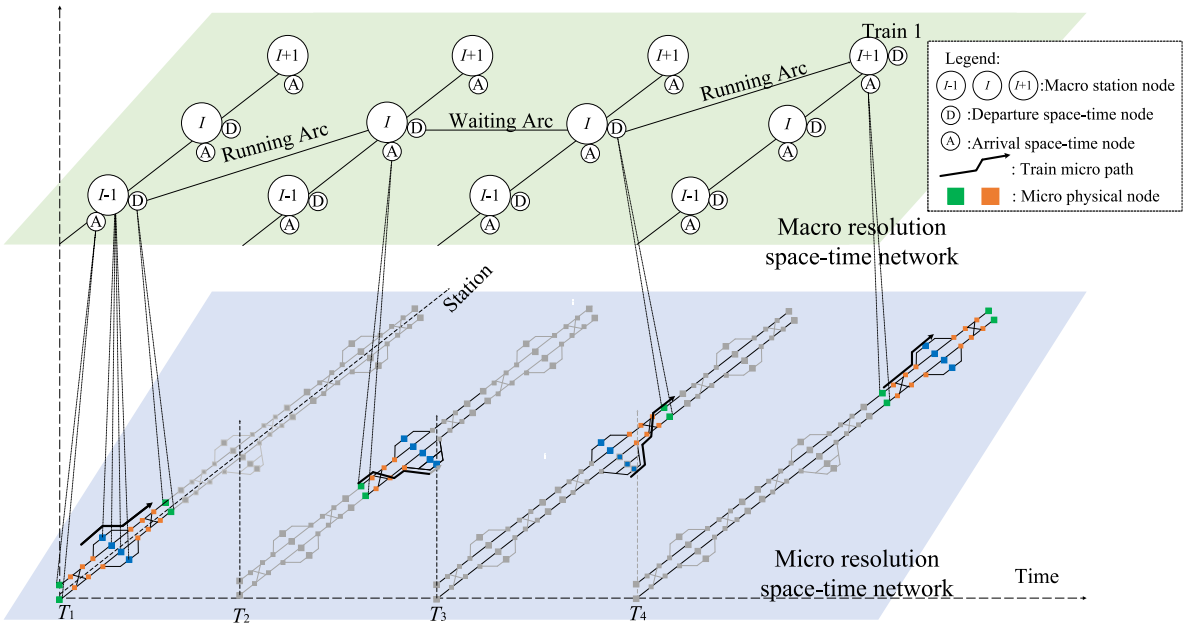


Fig. 6. Mapping relationship in multi-resolution layered space–time network.

- **Assumption 1**: For simplicity, each train is treated as a point in the network, where the train length is not taken into consideration. This assumption is widely adopted in the existing literature on train scheduling problems, e.g., Meng and Zhou (2014), Zhang et al. (2019b).
- **Assumption 2**: In order to avoid potential conflicts, a physical link at each station in the micro network can only be occupied by at most one train at any time. This is also the case in the real-world applications.
- **Assumption 3**: The trains traveling upstream and downstream pass through each segment in between two consecutive stations on the main tracks independently, while inside some stations, trains traveling upstream are allowed to use tracks dedicated for trains traveling downstream, and vice versa.

3.3. Coupled multi-resolution space–time network

The space–time network has been widely used in characterizing the rail transportation problems, such as cyclic train timetabling (Zhang et al., 2019c) and train maintenance planning (Luan et al., 2017; Zhang et al., 2020a). In this paper, we consider a coupled space–time network in two different resolutions. In the macro-resolution space–time network (termed by macro STN), each railway station is modeled as a series of space–time nodes with a large discrete time granularity (e.g., 1 min), while each turn-out or signal is constructed as a set of space–time nodes with a small time granularity (e.g., 1 s) in the micro-resolution network (termed by micro STN).

As shown in Fig. 6, the macro STN is denoted by $G^{SM} = \left(V^{SM}, A^{SM}\right)$, where V^{SM} and A^{SM} represent the sets of space–time nodes and arcs, respectively. Here, the discretized time horizon in the macro network is denoted by set $\mathcal{T}^M$, while T and T' denote the discrete time-stamps (i.e., $T, T' \in \mathcal{T}^M$). Then, notation (I, T) denotes a space–time node (i.e., $(I, T) \in V^{SM}$) and notation (I, J, T, T') denotes a space–time arc from (I, T) to (J, T') (i.e., $(I, J, T, T') \in A^{SM}$). Particularly, $(I, I, T, T' = T + 1)$ is a waiting arc at station I with the waiting event from time T to $T' = T + 1$. Then, a macro space–time path can be defined by a series of connected macro space–time nodes, e.g., $p^{SM} = \left\{\left(I-1, T_1\right), \left(I, T_2\right), \left(I, T_3\right), \left(I+1, T_4\right)\right\}$. Likewise, the micro STN is denoted by $G^{Sm} = \left(V^{Sm}, A^{Sm}\right)$ with space–time node $(i, t) \in V^{Sm}$ and space–time arc $(i, j, t, t') \in A^{Sm}$. Here, $\mathcal{T}^m$ is used to denote the discretized micro-resolution time horizon, and t and t' are discrete time-stamps (i.e., $t, t' \in \mathcal{T}^m$). For a waiting arc at track (i, j), one can use $(i, j, t, t' = t + 1)$ to denote the waiting event from time t to $t' = t + 1$ at departure–arrival tracks. For clarity, a micro path $p^{Sm} = \left\{\left(\mathrm{G}_0, t_1\right), \ldots, \left(\text{T-5}, t_7\right), \ldots, \left(\mathrm{G}_2, t_{12}\right)\right\}$ is illustrated in Fig. 7.

Coupling G^{SM} with G^{Sm} leads to the entire space–time network $G^S = (V^S, A^S)$, where $V^S = V^{SM} \cup V^{Sm}$ and $A^S = A^{SM} \cup A^{Sm}$. The node set V^S includes origin nodes, destination nodes, and intermediate nodes, while the arc set A^S consists of traveling arcs, waiting arcs, and dummy arcs if necessary. Node v^{SM} in the macro STN corresponds to a set of coupled nodes (denoted by $G^{Sm}(v^{SM})$) in the micro STN. This actually represents the mapping relationship between these two networks, as shown in Fig. 6. In the coupled network, a macro space–time path p^{SM} might correspond to multiple micro paths p^{Sm}. Meanwhile, due to the one-to-many coupling relationship between v^{SM} and $G^{Sm}(v^{SM})$, there are multiple micro paths p^{Sm} satisfying the coupling relationship for a given macro path p^{SM}, and the micro path can be updated or adjusted to a feasible solution without changing the macro path solution. In order

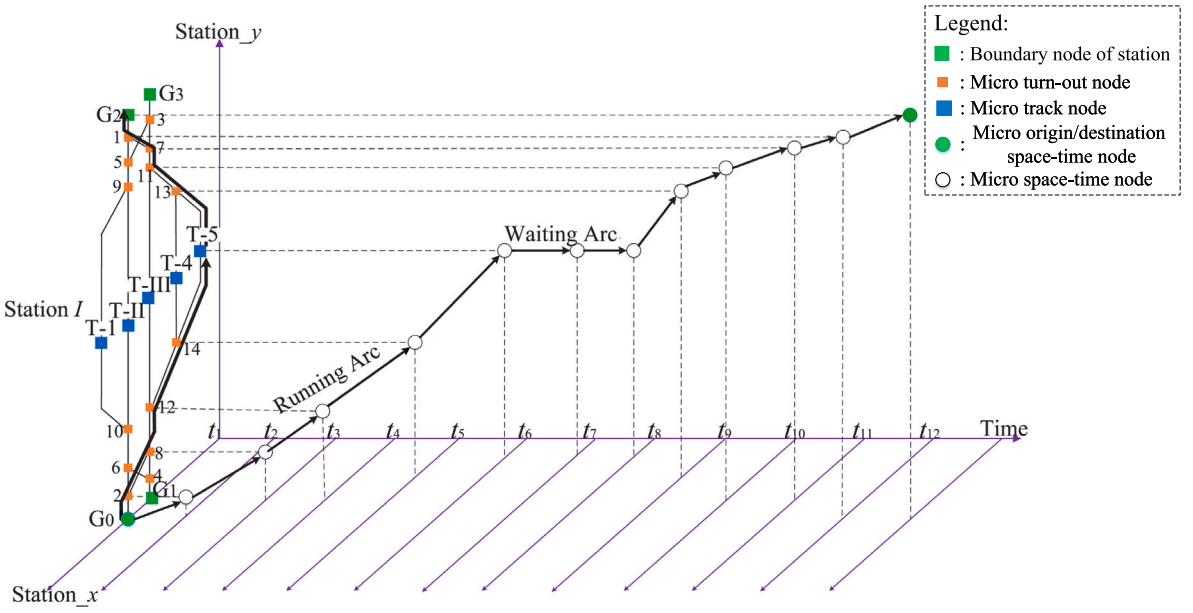


Fig. 7. Train routes in the micro-resolution space–time network.

to keep cost consistency across networks, macro arc cost $c(I,J,T,T')$ is regarded as the summation of its corresponding micro link costs $c(i,j,t,t')$ in the path finding process. Particularly, this study will not consider the waiting cost at starting and ending stations, i.e., the cost for waiting arcs at these two stations are set as zero. In general, a feasible joint optimization solution has the following properties, i.e., satisfying the requirements with respect to minimum headway, station capacity, and conflict-free routing.

4. Mathematical formulation

In this section, we intend to present a joint optimization model for the problem of interest in the multi-resolution space–time network. Next, the detailed formulation process will be elaborated, including the notations, parameters, system constraints, and objective function. In addition, the model complexity with respect to the decision variables and constraints is also analyzed in detail.

4.1. Notations

For modeling convenience, the notations used in the formulation are all listed in Table 2. Besides, to rigorously formulate our problem, we introduce three types of decision variables, including space–time arc selection variable, arrival resource occupation variable, and departure resource occupation variable, as summarized in Table 3. An illustration with respect to the arrival and departure variables is given in Fig. 9 for clarity.

4.2. System constraints and objective function

The mathematical model aims to generate the detailed train scheduling and routing schemes in the coupled multi-resolution space–time network. Its feasibility and consistency are guaranteed by a set of system constraints formulated as below, where the flow balance constraints (Group I) and the headway constraints (Group II) should be held in the both macro and micro resolution space–time networks. For clarity, we firstly discuss the macro constraints as the micro ones can be formulated similarly.

(1) System constraints

Group I: Flow balance constraints. As the scheduling process is to generate the space–time path for each train in the macro STN, flow balance constraints in this network should be imposed to keep the feasibility of the generated space–time path for each train $k \in K$ between its origin node $(I^o(k),T^o(k)) \in V^{SM}$ and destination node $\left(I^d(k),T^d(k)\right) \in V^{SM}$. Symbols $A^{SM^+}(I,T)$ and $A^{SM^-}(I,T)$ represent the outgoing link set and the ingoing link set of macro space–time node (I,T). We then formulate the flow balance constraints as follows.

$$\sum_{(J,T')\in A^{SM^+}(I,T)} x(k,I,J,T,T') - \sum_{(J,T')\in A^{SM^-}(I,T)} x(k,J,I,T',T) = \begin{cases} 1 & (I,T)=(I^o(k),T^o(k)) \\ -1 & (I,T)=\left(I^d(k),T^d(k)\right) \\ 0 & otherwise \end{cases} \quad \forall k \in K \tag{1}$$

Table 2
Definition of the involved indexes, sets and parameters.

Notations	Definition
Indexes	
Public indexes	
k	Index of trains
τ, τ'	Index of departure/arrival time-stamps
Macro indexes	
I, J	Index of macro physical nodes
(I, J)	Index of macro physical links
T, T'	Index of macro discrete time-stamps
(I, T)	Index of macro space–time nodes
(I, J, T, T')	Index of macro space–time arcs
a^S	Simplified index of macro space–time arc (I, J, T, T')
Micro indexes	
i, j	Index of micro physical nodes
(i, j)	Index of micro physical links
t, t'	Index of micro discrete time-stamps
(i, t)	Index of micro space–time nodes
(i, j, t, t')	Index of micro space–time arcs
a^s	Simplified index of micro space–time arc (i, j, t, t')
Sets	
Public sets	
N	Set of physical nodes $\left(N = N^M \cup N^m\right)$
A	Set of physical arcs $\left(A = A^M \cup A^m\right)$
V^S	Set of space–time nodes $\left(V^S = V^{SM} \cup V^{Sm}\right)$
A^S	Set of space–time arcs $\left(A^S = A^{SM} \cup A^{Sm}\right)$
K	Set of trains
Macro sets	
N^M	Set of macro physical nodes $\left(N^M \subseteq N\right)$
A^M	Set of macro physical links $\left(A^M \subseteq A\right)$
V^{SM}	Set of macro space–time nodes $\left(V^{SM} \subseteq V^S\right)$
A^{SM}	Set of macro space–time arcs $\left(A^{SM} \subseteq A^S\right)$
p^{SM}	Set of space–time nodes in a macro path
$\Psi\left(a^S\right)$	Set of incompatible arcs with respect to macro arc a^S
$\mathcal{T}^M$	Set of macro discrete time-stamps, $T \in \mathcal{T}^M$
Micro sets	
N^m	Set of micro physical nodes $(N^m \subseteq N)$
A^m	Set of micro physical links $(A^m \subseteq A)$
V^{Sm}	Set of micro space–time nodes $\left(V^{Sm} \subseteq V^S\right)$
A^{Sm}	Set of micro space–time arcs $\left(A^{Sm} \subseteq A^S\right)$
p^{Sm}	Set of space–time nodes in a micro path
$\Psi\left(a^s\right)$	Set of incompatible arcs with respect to micro arc a^s
$\mathcal{T}^m$	Set of micro discrete time-stamps, $t \in \mathcal{T}^m$
Coupling sets	
$G^{Sm}(v^{SM})$	Set of coupling micro nodes for macro node v^{SM}
Parameters	
Macro parameters	
$c(k, I, J, T, T')$	Traveling cost of macro space–time link (I, J, T, T') for train k
$h^A(I, J)$	Minimum arrival headway for macro physical link (I, J)
$h^D(I, J)$	Minimum departure headway for macro physical link (I, J)
$I^o(k)$	Macro origin physical node of train k
$I^d(k)$	Macro destination physical node of train k
$T^o(k)$	Macro origin time-stamp of train k
$T^d(k)$	Macro destination time-stamp of train k
Micro parameters	
$h^A(i, j)$	Minimum arrival headway for micro physical link (i, j)
$h^D(i, j)$	Minimum departure headway for micro physical link (i, j)
$i^o(k)$	Micro origin physical node of train k
$i^d(k)$	Micro destination physical node of train k
$t^o(k)$	Micro origin time-stamp of train k
$t^d(k)$	Micro destination time-stamp of train k

Table 3
Definition of decision variables.

Notations	Definition
Macro decision variables	
$x(k,I,J,T,T')$	Indicator variable for space–time arc selection in the macro space–time network, = 1 if train k uses the macro space–time arc (I,J,T,T'); = 0 otherwise.
$d(k,I,J,T)/d(k,I,J,\tau)$	Indicator variable for macro departure resource occupation, = 1 if train k uses the departure resource of the macro link (I,J) at time T/τ; = 0 otherwise.
$a(k,I,J,T')/a(k,I,J,\tau')$	Indicator variable for macro arrival resource occupation, = 1 if train k uses the macro arrival resource of the link (I,J) at time T'/τ'; = 0 otherwise.
Micro decision variables	
$x(k,i,j,t,t')$	Indicator variable for space–time arc selection in the micro space–time network, = 1 if train k uses the micro space–time arc (i,j,t,t'); = 0 otherwise.
$d(k,i,j,t)/d(k,i,j,\tau)$	Indicator variable for micro departure resource occupation, = 1 if train k uses the departure resource of the micro link (i,j) at time t/τ; = 0 otherwise.
$a(k,i,j,t')/a(k,i,j,\tau')$	Indicator variable for micro arrival resource occupation, = 1 if train k uses the micro arrival resource of the link (i,j) at time t'/τ'; = 0 otherwise.

As for the train routing problem in the micro STN, its flow balance constraints can be formulated similarly for each train k, according to its origin $(i^o(k),t^o(k))$ and destination $(i^d(k),t^d(k))$ in the micro STN.

$$\sum_{(j,t')\in A^{Sm+}(i,t)} x(k,i,j,t,t') - \sum_{(j,t')\in A^{Sm-}(i,t)} x(k,j,i,t',t) = \begin{cases} 1 & (i,t)=(i^o(k),t^o(k)) \\ -1 & (i,t)=\left(i^d(k),t^d(k)\right) \\ 0 & otherwise \end{cases} \quad \forall k\in K \tag{2}$$

Group II: Headway constraints. The minimum headways $h^A(I,J)$ and $h^D(I,J)$ should be guaranteed to comply the safety distance requirement on each physical segment (I,J). In this paper, we use the symbol $\Psi(a^{SM})$ to represent the incompatible set with respect to arc $a^{SM}=(I,J,T,T')$. That is, when train $k\in K$ uses link (I,J) in time window $[T,T']$, all the space–time arcs starting from node I in time interval $\left[T,T+h^D(I,J)\right)$ or terminating at J in time interval $\left[T',T'+h^A(I,J)\right)$ should belong to set $\Psi(a^{SM})$. Its formal definition is given as below.

$$\Psi\left(a^{SM}\right)=\left\{\left(I,J,\tau,\tau'\right)|0\le(\tau-T)<h^D(I,J),0\le\left(\tau'-T'\right)<h^A(I,J)\right\}\ \forall a^{SM}=(I,J,T,T')\in A^{SM} \tag{3}$$

Fig. 8 gives an illustration of the incompatible set (the green shadow area) with respect to arc $(I,J,3,4)$ under a 2-minute headway. The corresponding set can be represented by

$$\Psi((I,J,3,4))=\{(I,J,1,4),(I,J,2,4),(I,J,2,5),(I,J,3,4),(I,J,3,5),(I,J,3,6),(I,J,4,5),(I,J,4,6),(I,J,4,7)\}.$$

If space–time arc $(I,J,3,4)$ is used, any arcs overlapping with its incompatible set cannot be selected by other trains.

We then formulate the headway constraint on each segment based on the incompatible set representation, given below.

$$\sum_{k\in K}\sum_{(I,J,\tau,\tau')\in\Psi((I,J,T,T'))} x\left(k,I,J,\tau,\tau'\right)\le 1 \quad \forall(I,J,T,T')\in A^{SM} \tag{4}$$

Theoretically, we need to enumerate a corresponding set $\Psi((I,J,T,T'))$ for each occupied space–time arc (I,J,T,T'), which may cause memory issues in solving the model. Therefore, this paper proposes two new decision variables, i.e., the departure resource occupation variable $d(k,I,J,\tau)$ and the arrival resource occupation variable $a(k,I,J,\tau')$ to reformulate the aforementioned headway constraints. All resource occupation variables should be set to 0 initially, and then comply the following constraints (5) and (6).

$$d\left(k,I,J,\tau\right)=1 \quad \forall\tau\in\left[T,T+h^D(I,J)\right),(I,J,T,T')\in A^{SM},k\in K \tag{5}$$

$$a\left(k,I,J,\tau'\right)=1 \quad \forall\tau'\in\left[T',T'+h^A(I,J)\right),(I,J,T,T')\in A^{SM},k\in K \tag{6}$$

As shown in Fig. 9, if train k is scheduled to use arc (I,J,T,T'), it is clear that departure and arrival resources corresponding to the incompatible set are not available for other trains, and the resource occupation variables within headway ranges (i.e., $h^A(I,J)$ and $h^D(I,J)$) should be also marked as 1.

Following the definition of resource occupation variables, incompatible-arc-based constraints (4) can be reformulated as (7)~(10) by using resource occupation variables.

$$\sum_{k\in K} d\left(k,I,J,T\right)\le 1 \quad \forall(I,J)\in A^M,T\in\mathcal{T}^M \tag{7}$$

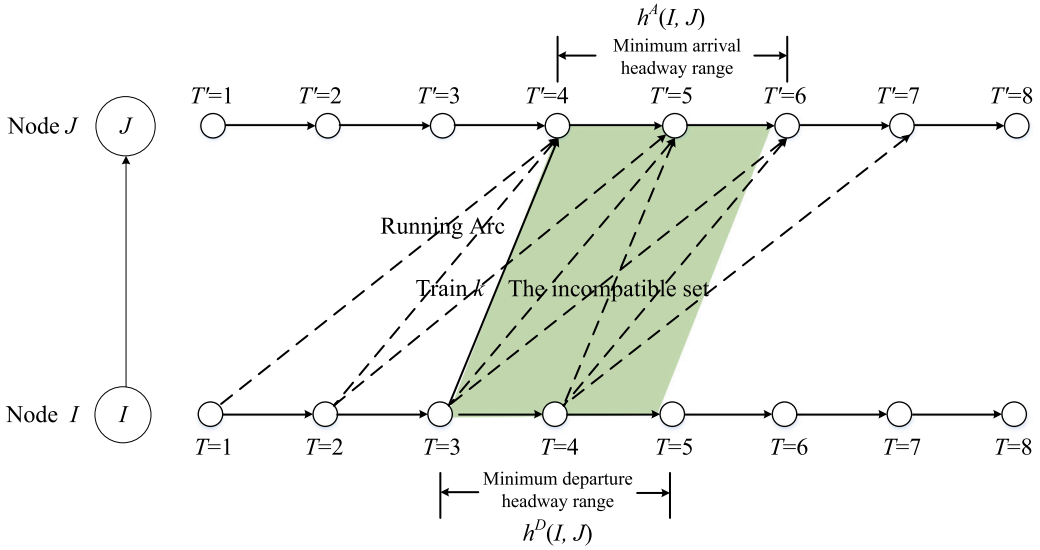


Fig. 8. Illustration for the incompatible set.

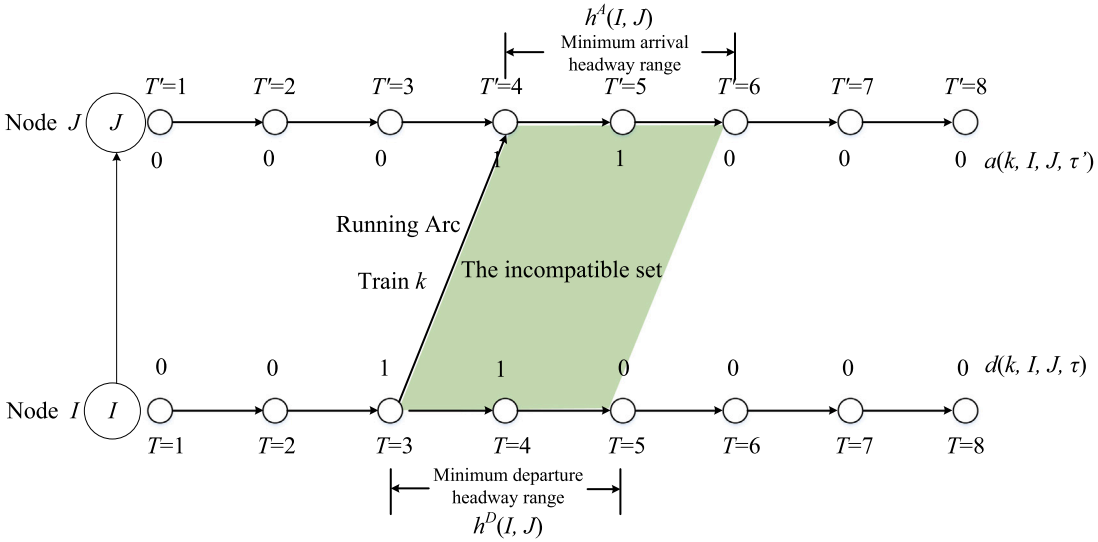


Fig. 9. Illustration of the incompatible set and resource occupation variables.

$$\sum_{k\in K} a\left(k,I,J,T'\right) \le 1 \quad \forall (I,J) \in A^M, T' \in \mathcal{T}^M \tag{8}$$

$$\sum_{T'\in\mathcal{T}^M} x(k,I,J,T,T') = d(k,I,J,\tau) \quad \forall \tau \in \left[T, T + h^D(I,J)\right), (I,J) \in A^M, T \in \mathcal{T}^M, k \in K \tag{9}$$

$$\sum_{T\in\mathcal{T}^M} x(k,I,J,T,T') = a(k,I,J,\tau') \quad \forall \tau' \in \left[T', T' + h^A(I,J)\right), (I,J) \in A^M, T' \in \mathcal{T}^M, k \in K \tag{10}$$

For the train routing problem in the micro STN based on the safety clearance time of the turn-outs, its headway constraints can be reformulated in a similar way.

$$\sum_{k\in K} d\left(k,i,j,t\right) \le 1 \quad \forall (i,j) \in A^m, t \in \mathcal{T}^m \tag{11}$$

$$\sum_{k\in K} a\left(k,i,j,t'\right) \le 1 \quad \forall (i,j) \in A^m, t' \in \mathcal{T}^m \tag{12}$$

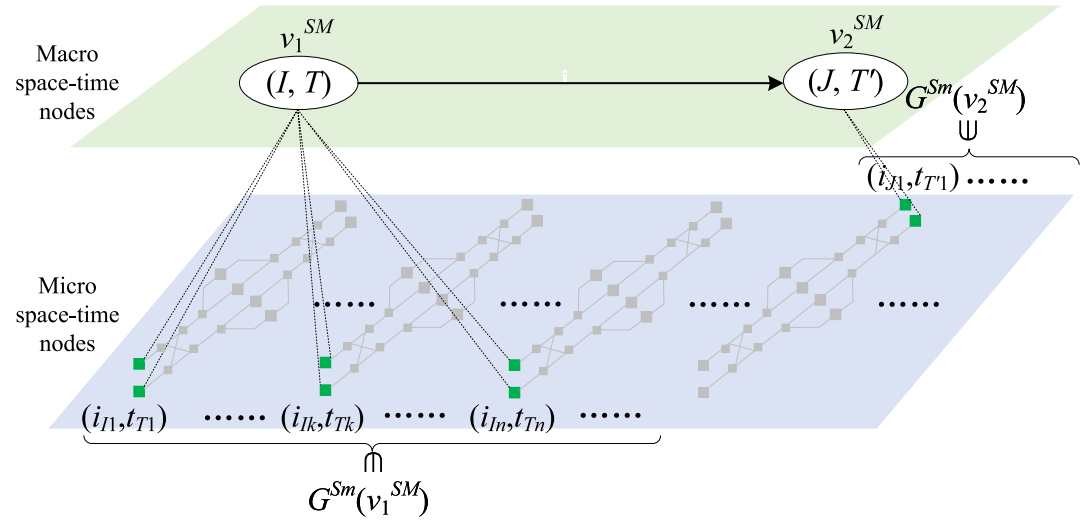


Fig. 10. An illustration of coupling set $G^{Sm}(v^{SM})$.

$$\sum_{t' \in \mathcal{T}^m} x(k,i,j,t,t') = d(k,i,j,\tau) \quad \forall \tau \in \left[t, t + h^D(i,j)\right), (i,j) \in A^m, t \in \mathcal{T}^m, k \in K \tag{13}$$

$$\sum_{t \in \mathcal{T}^m} x(k,i,j,t,t') = a(k,i,j,\tau') \quad \forall \tau' \in \left[t', t' + h^A(i,j)\right), (i,j) \in A^m, t' \in \mathcal{T}^m, k \in K \tag{14}$$

With constraints (1)~(2), (7)~(14), we can optimize the macro scheduling and micro routing problems separately in different space–time networks. The consistency and coupling constraints between the macro and micro STNs are also needed to ensure the feasibility of the joint optimization solutions, which will be discussed in the following parts.

Group III: Space–time network coupling constraints. According to the coupling structure between the macro and micro STNs, if train k is scheduled to use a macro arc $a^{SM} = (I, J, T, T') \in A^{SM}$ (denote $v_1^{SM} = (I, T)$ and $v_2^{SM} = (J, T')$), there exist at least two micro space–time nodes v_1^{Sm}, v_2^{Sm} coupled with v_1^{SM}, v_2^{SM}, respectively. We represent the coupling relationship between a macro space–time node v^{SM} and its coupled micro space–time nodes by set $G^{Sm}(v^{SM})$, i.e., any micro space–time nodes coupled with macro space–time nodes $v^{SM} = (I, T)$ should belong to set $G^{Sm}(v^{SM})$. For example $G^{Sm}(v_1^{SM}) = G^{Sm}((I,T)) = \{(i_{I1}, t_{T1}), \ldots, (i_{Ik}, t_{Tk}), \ldots, (i_{In}, t_{Tn})\}$ in Fig. 10, where the number of the elements of $G^{Sm}(v^{SM})$ is determined by the multiple between two time resolutions. On the contrary, if a macro space–time arc a^{SM} is not used, no micro space–time arc starting with nodes in $G^{Sm}(v_1^{SM})$ or ending with nodes in $G^{Sm}(v_2^{SM})$ can be used (see Eqs. (15) and (16)). Then, the micro paths that satisfy the headway constraints (Eqs. (11)~(14)) and the following coupling constraints (Eqs. (15) and (16)) are feasible in the train routing problem.

$$x\left(k,I,J,T,T'\right) = \sum_{(i,t) \in G^{Sm}((I,T))} \sum_{(j,t') \in A^{Sm+}(i,t)} x(k,i,j,t,t') \quad \forall \left(I,J,T,T'\right) \in A^{SM}, k \in K \tag{15}$$

$$x\left(k,I,J,T,T'\right) = \sum_{(j,t') \in G^{Sm}((J,T'))} \sum_{(i,t) \in A^{Sm-}(j,t')} x(k,i,j,t,t') \quad \forall \left(I,J,T,T'\right) \in A^{SM}, k \in K \tag{16}$$

Group IV. Decision variable constraints. All the space–time decision variables should be binary variables, stated in the following constraints.

$$x(k,I,J,T,T') \in \{0,1\} \quad \forall (I,J,T,T') \in A^{SM}, k \in K \tag{17}$$

$$d(k,I,J,T) \in \{0,1\} \quad \forall (I,J,T,T') \in A^{SM}, k \in K \tag{18}$$

$$a(k,I,J,T') \in \{0,1\} \quad \forall (I,J,T,T') \in A^{SM}, k \in K \tag{19}$$

$$x(k,i,j,t,t') \in \{0,1\} \quad \forall (i,j,t,t') \in A^{Sm}, k \in K \tag{20}$$

$$d(k,i,j,t) \in \{0,1\} \quad \forall (i,j,t,t') \in A^{Sm}, k \in K \tag{21}$$

$$a(k,i,j,t') \in \{0,1\} \quad \forall (i,j,t,t') \in A^{Sm}, k \in K \tag{22}$$

Table 4
The number of decision variables and constraints.

Variables or constraints	Macroscope	Microscope
Decision variables		
$x(k,I,J,T,T')/x(k,i,j,t,t')$	$\lvert K\rvert \cdot \lvert A^{SM}\rvert$	$\lvert K\rvert \cdot \lvert A^{Sm}\rvert$
$a\left(k,I,J,T'\right)/a\left(k,i,j,t'\right)$	$\lvert K\rvert \cdot \lvert A^{M}\rvert \cdot \lvert T\rvert$	$\lvert K\rvert \cdot \lvert A^{m}\rvert \cdot \lvert T\rvert \cdot T^{Mm}$
$d\,(k,I,J,T)/d\,(k,i,j,t)$	$\lvert K\rvert \cdot \lvert A^{M}\rvert \cdot \lvert T\rvert$	$\lvert K\rvert \cdot \lvert A^{m}\rvert \cdot \lvert T\rvert \cdot T^{Mm}$
Constraints		
Group I: (1)–(2)	$\lvert K\rvert \cdot \lvert N^{M}\rvert \cdot \lvert T\rvert$	$\lvert K\rvert \cdot \lvert N^{m}\rvert \cdot \lvert T\rvert \cdot T^{Mm}$
Group II: (7)–(8), (11)–(12)	$\lvert A^{M}\rvert \cdot \lvert T\rvert$	$\lvert A^{m}\rvert \cdot \lvert T\rvert \cdot T^{Mm}$
Group II: (9)–(10), (13)–(14)	$\lvert K\rvert \cdot \sum_{(I,J)\in A^{M}} \lvert h(I,J)\rvert \cdot \lvert T\rvert$	$\lvert K\rvert \cdot \sum_{(i,j)\in A^{m}} \lvert h(i,j)\rvert \cdot \lvert T\rvert \cdot T^{Mm}$
Group III: (15)–(16)	$\lvert K\rvert \cdot \lvert A^{SM}\rvert$	

(2) Objective function

In the multi-resolution space–time network model, the cost of a feasible macro path is equivalent to that of its corresponding micro path as a result of consistency properties. In other words, we only need to consider the objective function at the macro level, i.e.,

$$\min \quad Z = \sum_{k\in K} \sum_{(I,J,T,T')\in A^{SM}} c(k,I,J,T,T') \cdot x(k,I,J,T,T'). \tag{23}$$

4.3. Mathematical model

According to Eqs. (1)∼(23), a joint optimization model (M1) of the train scheduling and routing problem can be formulated as below:

$$\text{M1:} \begin{cases} \min \quad Z = \sum_{k\in K} \sum_{(I,J,T,T')\in A^{SM}} c(k,I,J,T,T') \cdot x(k,I,J,T,T') \\ \text{s.t.} \quad \text{constraints} \quad (1)\text{–}(2),\ (7)\text{–}(22) \end{cases}. \tag{24}$$

4.4. Complexity analyses of the proposed model

The scale of the proposed ILP model is proportional to the numbers of nodes, arcs, trains, and time resolutions, and the total numbers of decision variables and constraints are summarized in Table 4, where the symbol $|\cdot|$ denotes the cardinality of a set and the parameter T^{Mm} represents the multiple between two time resolutions. For example, the parameter T^{Mm} should be set as 60 if the time resolutions are set to 1 min (=60 s) and 1 s in the macro and micro STNs respectively.

5. Solution algorithms

As discussed in previous sections, the joint optimization needs to consider both macro and micro information in the coupled network, which will be a very large-scale problem if the micro time granularity is set as a small value. With the multi-resolution network representation, the primal problem can be decomposed into two sub-problems: one is the macro scheduling problem of determining the train departure and arrival times at each station, and the other is the micro routing problem of determining specific routes inside each station. However, it is usually difficult to find a feasible routing scheme in a busy rail corridor when a train schedule is pre-given due to the existence of the coupling relationship. Therefore, we shall design an iterative solution process based on the branch-and-bound searching framework. Specifically, at each node of the searching tree, an effective search procedure is proposed to solve the involved models, and the computation results can be used for branching in the following search. At the beginning of the search process (i.e., root node), the train scheduling problem is firstly solved to generate the optimal schedule; then, this solution will be passed into the micro train routing process, and the detailed routing scheme inside each station needs to be found. If no feasible route exists, the conflicting information will be used to reformulate the current model, with which different branches can be produced. The above search goes on iteratively until the termination criteria are met, and the best encountered feasible solution is outputted as the near-optimal solution to the primal model. Next, we shall address the detailed techniques in the algorithm design.

5.1. Solution algorithms for macro and micro problems

5.1.1. Problem decomposition

The primal mathematical model (M1) is firstly decomposed into a macro train scheduling model (M2) and a corresponding routing process in the micro STN (M3), as shown in Fig. 11. In the searching process, model M2 related to the train scheduling process can be solved in the macro STN by using the LR, DP and priority rule (PR) methods. Then, the routing process M3 will be

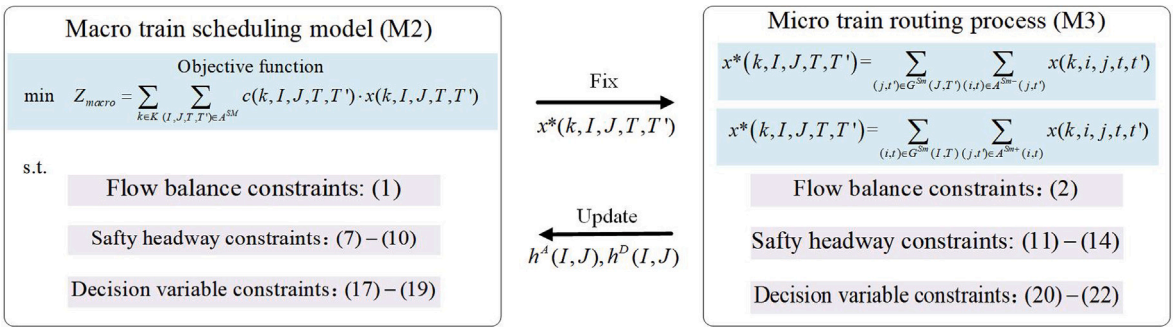


Fig. 11. Model decomposition of the problem.

solved with the fixed macro decision variables $x(k,I,J,T,T')$ generated by model M2. If no feasible micro routes can be found, some customized extra constraints should be further imposed on model M2, based on the adjustment of headway parameters $h^A(I,J)$ and $h^D(I,J)$ (see Section 5.3).

For those selected macro arcs $a^{SM} = \{(k,I,J,T,T') | x(k,I,J,T,T') = 1\}$, we can generate the corresponding coupled micro space–time node sets (i.e., $G^{Sm}((I,T))$ and $G^{Sm}((J,T'))$) for each train and station. Then, based on the generated train schedule, we solve the micro train routing process M3 based on these fixed departure and arrival events at station boundaries by using LR and DP.

5.1.2. Macro space–time path finding algorithm

(1) Lagrangian relaxation

Model M2 and routing process M3 can be viewed as multi-train space–time routing problems with headway constraints. For descriptive convenience, we firstly use the LR algorithm to solve the problem M2 at the macro level. In this model, train safety headway constraints (7) and (8) are actually hard constraints, which can be dualized via Lagrangian multipliers $\lambda(I,J,T) \geq 0$ and $\mu(I,J,T') \geq 0$. Then, we obtain the relaxed model of M2, shown in Eqs. (25) and (26).

$$\begin{aligned} \min \quad & Z = \sum_{k\in K} \sum_{(I,J,T,T')\in A^{SM}} c(k,I,J,T,T') \cdot x(k,I,J,T,T') \\ & + \sum_{(I,J,T)} \lambda(I,J,T) \left[\sum_{k\in K} d(k,I,J,T) - 1 \right] + \sum_{(I,J,T')} \mu(I,J,T') \left[\sum_{k\in K} a(k,I,J,T') - 1 \right] \end{aligned} \tag{25}$$

$$\text{s.t.} \begin{cases} \text{Flow balance constraints: (1)} & (a) \\ \text{Resource occupation variable constraints: (9)–(10)} & (b) \\ \text{Decision variable constraints: (17)–(19)} & (c) \end{cases} \tag{26}$$

By substituting Eq. (26)b into objective function Eq. (25) and regrouping the variables, we have a more systematic view of the components in the relaxed model, i.e., Eqs. (27) and (28), where the generalized cost is computed by $c'(k,I,J,T,T') = c(k,I,J,T,T') + \lambda(I,J,T) + \mu(I,J,T')$.

$$\min \quad Z = \sum_{k\in K} \sum_{(I,J,T,T')\in A^{SM}} c'(k,I,J,T,T') \cdot x(k,I,J,T,T') - \sum_{(I,J,T)} \lambda(I,J,T) - \sum_{(I,J,T')} \mu(I,J,T') \tag{27}$$

$$\text{s.t.} \begin{cases} \text{Flow balance constraints: (1)} \\ \text{Decision variable constraints: (17)–(19)} \end{cases} \tag{28}$$

Denote this relaxed model as M4. Then, it can be further decomposed into a series of sub-problems for scheduling individual trains, which can be effectively solved by DP in parallel since each sub-problem contains only the flow balance constraints. Note that the optimal objective of the relaxed model is a lower bound to the macro train scheduling model. To find the tightest lower bound, we then solve the Lagrangian dual problem via a sub-gradient algorithm. Let λ^0, μ^0 be the initial value of the Lagrangian multipliers, and the subsequent values λ^{iter} and μ^{iter} for $iter = 1,2,3,\ldots$ can be updated based on Eqs. (29) and (30), where $x(k,I,J,T,T')$ is the optimal solution of the relaxed model when $\lambda(I,J,T) = \lambda^{iter}(I,J,T), \mu(I,J,T') = \mu^{iter}(I,J,T')$, and $\theta^{iter} = \frac{1}{iter+1}$ is the sub-gradient step size in iteration $iter$.

$$\lambda^{iter+1}(I,J,T) = \max \left\{ 0, \lambda^{iter}(I,J,T) + \theta^{iter} \left(\sum_{k\in K} \sum_{T'\in \mathcal{T}^M} x(k,I,J,T,T') - 1 \right) \right\}, \forall (I,J) \in A^M, T \in \mathcal{T}^M \tag{29}$$

$$\mu^{iter+1}(I,J,T') = \max \left\{ 0, \mu^{iter}(I,J,T') + \theta^{iter} \left(\sum_{k\in K} \sum_{T\in \mathcal{T}^M} x(k,I,J,T,T') - 1 \right) \right\}, \forall (I,J) \in A^M, T' \in \mathcal{T}^M \tag{30}$$

(2) Dynamic programming in time-dependent shortest path problems

When the relaxed model M4 is decomposed into a sequence of sub-problems with respect to each train, each sub-problem from M4 can be viewed as a time-dependent shortest path problem in the macro STN, which can be solved by a DP algorithm efficiently. Up to now, DP has been widely used in multiple-stage decision-making problems. Compared with other shortest path algorithms, DP algorithm can be extended easily to find time-dependent shortest paths (Ziliaskopoulos and Mahmassani, 1993; Mahmoudi et al., 2019; Shang et al., 2019, 2021). In this section, the time-dependent shortest path problem in the macro STN can be solved by DP efficiently (Algorithm 1), which then leads to the exact optimum for the relaxed model.

Algorithm 1 Dynamic programming for solving macro Lagrangian relaxed model M4

Input:

Railway space-time network $G^{SM} = (V^{SM}, A^{SM})$; Planning time horizon $\mathcal{T}^M$; space-time origin/destination node $(I^o(k), T^o(k))/(I^d(k), T^d(k))$ of each train k.

Output:

Space-time path with the minimum cost for each train k.

Step 1 (Initialization):
for each physical railway link $l = (I, J)$ **do**
  for each time-stamp $T, T' \in \mathcal{T}^M$ **do**
    Initialize Lagrangian multipliers $\lambda(I, J, T), \mu(I, J, T')$
  end for
end for
Step 2 (Updating label cost dynamically):
for each train $k \in K$ **do**
  for each space-time node (I, T) **do**
    Initialize physical node predecessor array: $node_pred[I, T] = -1$
    Initialize time predecessor array: $time_pred[I, T] = -1$
    Initialize label cost array: $label_cost[I, T] = 9999$
  end for
  for each time-stamp $T \in \mathcal{T}^M$ **do**
    for each physical railway node $I \in N^M$ **do**
      for each outgoing link $l = (I, J)$ of current node I **do**
        I:= from node ID of link l
        J:= to node ID of link l
        TT:= travel time of link l
        $T' := T + TT$
        Compute the $general_cost$ of space time arc (I, J, T, T')
        $label_cost_temp = label_cost[I, T] + general_cost$
        if $label_cost_temp < label_cost\ [J, T']$ **then**
          Update the predecessor node:
          $node_pred[J, T'] = I$
          $time_pred[J, T'] = T$
          $label_cost[J, T'] = label_cost_temp$
        end if
      end for
    end for
  end for
end for
Step 3 (Backtracing):
For each train k, using the predecessor arrays $node_pred, time_pred$, and $label_cost$ to trace back from the space-time destination node $(I^d(k), T^d(k))$ to origin node $(I^o(k), T^o(k))$. We then can obtain the routing information for each train and the corresponding objective value.

(3) Priority rule (PR) for upper bound solutions

When the sub-gradient algorithm can finally find a feasible train schedule in the macro STN, this solution must be the optimal solution to the primal macro scheduling problem. Note that it is difficult to obtain a feasible train schedule by using the sub-gradient algorithm to solve the Lagrangian dual problem. When it returns an infeasible solution, we then call this solution as a Lagrangian lower bound solution. Next, we need to design the priority rule to transform this Lagrangian lower bound solution to a feasible macro schedule, in order to update the upper bound in the solution process. In detail, we use the priority rule to reschedule trains in the lower bound solution based on the resource occupation variable representation (see Algorithm 2).

5.1.3. Micro space–time path finding algorithm

When an optimal train schedule is generated by the priority rule, we then solve the specific train routing process M3 in the micro network for each railway station in parallel given train arrival and departure times. Similar to the macro train scheduling algorithm,

Algorithm 2 Priority rule to generate a feasible schedule

Input:
Railway space-time network $G^{SM} = (V^{SM}, A^{SM})$, the Lagrangian lower bound solution
Output:
A feasible schedule (i.e., an upper bound solution)
Step 1 (Computing Lagrangian profits):
Compute the Lagrangian profit for each train k in the lower bound solution (i.e., the optimal generalized cost of each sub-problem).
Step 2 (Sorting trains):
Reorder the indexes of trains as $k'_1, k'_2, ..., k'_{|K|}$ according to the branching rules of the current BB node (firstly) and the descending order of Lagrangian profits (secondly).
Step 3 (Rescheduling):
for each train k' in sequence $k'_1, k'_2, ..., k'_{|K|}$ **do**
    Step 3.1: Find the space-time shortest path $p^{k'}$ by DP (Algorithm 1) in network G^{SM} without the marked resources for train k'.
    Step 3.2: Mark the space-time occupation resources and related headway resources of path $p^{k'}$ (The marked resources cannot be occupied by any other subsequent trains again).
end for

we can dualize the headway constraints in process M3 with LR procedure and apply DP to solve the micro path for each train inside every station. If a feasible solution to routing process M3 is found, we will consider the corresponding scheduling and routing scheme as a feasible solution to the primal model. Otherwise, the PR is still needed to generate micro train route solutions in each iteration. If all the generated micro solutions are still not feasible to routing process M3, there exists at least one conflicting train pair which should be identified to proceed the following branch-and-bound procedure. Then, the conflicting information represented by a set of constraints, will be fed back to the train scheduling model. Afterwards, each branching node will be solved repeatedly until a feasible solution is obtained for the joint optimization problem.

5.2. Branch-and-bound procedure

Next, we will develop a branch-and-bound algorithm to decompose the solution space of the original problem into different branches by implicitly enumerating possible train priority pairs.

To the best of our knowledge, trains on a double-track corridor are usually scheduled separately in each direction, where the feasibility of micro routes at each station is not taken into account explicitly. In this case, conflicts between trains traveling in the opposite directions often occur at station junctions when the timetable is made to the real operations. As shown in Fig. 2(c), due to the flexible use of departure–arrival tracks (1, 4 and 5), trains k_m and k_n will conflict at station throat A at the micro level. Before performing the constraint updating procedure, two types of train conflicts should be considered, which are the conflicts of trains traveling in the same direction and the conflicts of trains traveling in the opposite directions. For conflicting pairs in the same direction, we adopt the First Come First Serve (FCFS) strategy, which can avoid the excessive size of branched nodes. For those in the opposite directions, it is necessary to schedule passing sequences or priorities at the station junction to improve the resource utilization in the junction region. If multiple optimal solutions exist with the same objective value, we will choose one solution with the minimum duration of operation time horizon as the final optimal solution.

We first give a general description of the branch-and-bound procedure along with the overall framework of the algorithm illustrated in Fig. 12.

r, m, n: the search node indexes;
ANL: the active search node list;
$y(I, k_m, k_n)$: the branching variable of trains k_m and k_n.

Step 1: (Initialization). Create a root node $r = 1$, which represents the primal problem. Insert this node into the active node list ANL.

Step 2: (Scheduling and routing). Select an active BB node from ANL based on a given node selection rule (e.g., depth-first search and breadth-first search), solve the macro subproblem M2 and the micro routing process M3 (see Section 5.1).

Step 3: (Upper bound updating). The Lagrangian LB of the macro train scheduling problem is recorded in current active BB node for the bounding process. If a feasible solution is found for the joint "macro–micro" optimization problem in step 2, the global UB will be updated based on the minimum of the current global UB and the newly encountered objective value.

Step 4: (Stopping criterion and node selection). The current active BB node will be removed after updating the global UB in step 3. When the ANL list becomes empty, this algorithm will terminate and output the solution(s) with the best objective value on all generated feasible BB nodes.

Step 5: (Branching). Identify all the conflicting train pairs at each station. For each conflicting train pair (k_m, k_n) inside station I, enumerate two branches explicitly (train k_m first or train k_n first) in the BB algorithm. Then, create two new search nodes m and n with the corresponding branches and push back nodes into the list ANL.

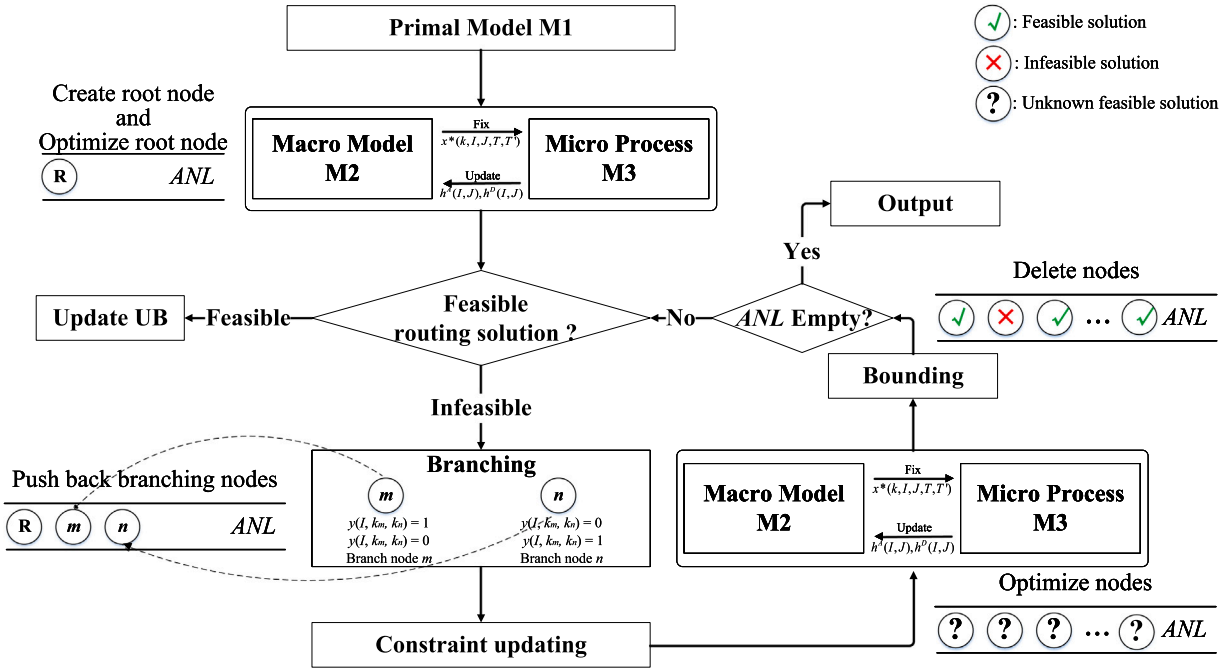


Fig. 12. The overall solution framework.

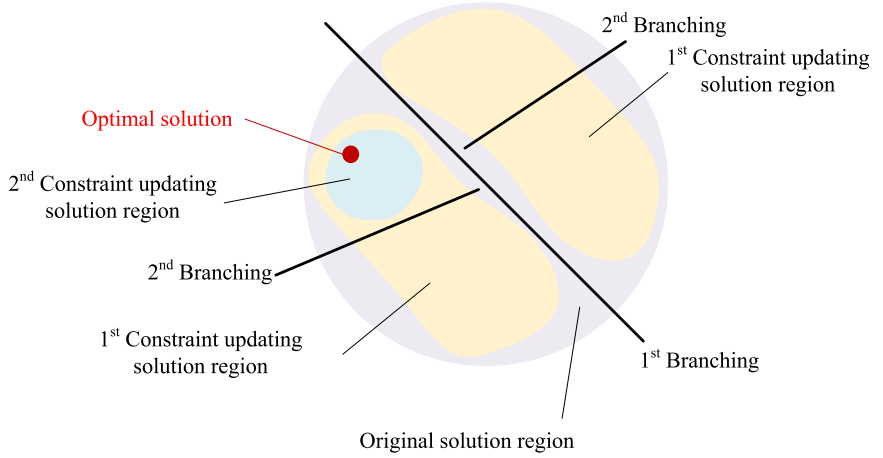


Fig. 13. Iterative process of solution.

Step 6: (Constraint updating). Based on branching rules in Section 5.2.1, update the alternative headway set ($h^A(I,J)$ and $h^D(I,J)$) by constraint updating (CU) procedures. Go back to Step 2.

Next, we illustrate the BB by the change of solution region in Fig. 13. Specifically, the original solution region can be decomposed into two sub-regions (i.e., two child nodes of the root node) which can be solved in parallel in the 1st branching procedure. Then, the decomposed regions can be further updated through each CU procedure (i.e., the 1st constraint updating solution region). After processing the BB and CU for multiple times, the optimal solution will be restricted in a small region which can be found easier.

5.2.1. Branching rules

As for the strategy of resolving conflicts between trains traveling in the opposite directions, one train can be scheduled to pass through the conflict block first, while the other train can choose an appropriate time to pass after the block is cleared. The proposed branching rule for determining priorities can be found similarly in Zhou and Zhong (2005), Zhou and Zhong (2007), Wu et al. (2021). Due to many uncertain situations in the resource utilization of adjacent stations, the two branching decisions (train k_m first or train k_n first) usually produce different train schedule structures. For example, branch A (train k_m first) will occupy segment

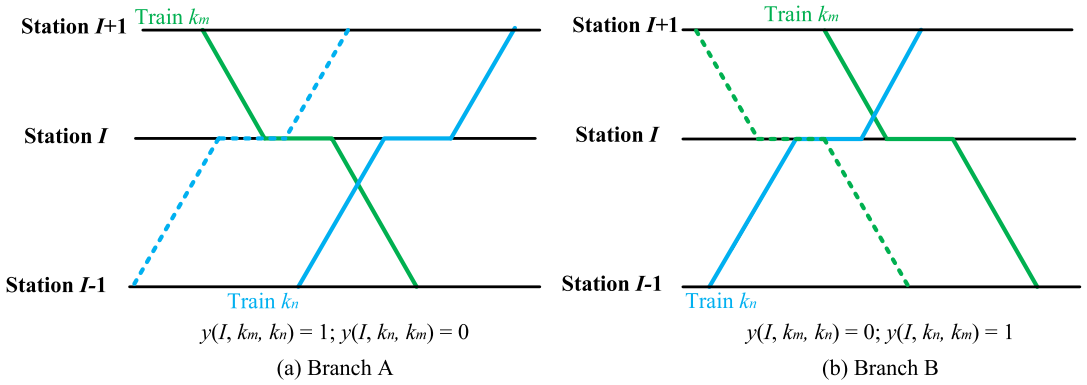


Fig. 14. Branching scheme at station I.

$(I, I+1)$ to resolve conflicts, while branch B (train k_n first) will occupy segment $(I-1, I)$ in Fig. 14. It should be pointed out that the local rescheduling or adjustment is easy only with a small number of conflicts. When there are many conflicts in the overall train schedule, any adjustments on the train pair priority will not only affect the local resource utilization scheme, but also cause secondary delay for other trains (Zhou and Zhong, 2005, 2007). Therefore, it is necessary to find the optimal priority combination to minimize the delay as much as possible. In practice, when multiple conflicts occur simultaneously in a train schedule, the conflicting train pair with the earliest occurrence time will be selected in priority to generate BB nodes and update constraints to avoid large BB node enumerations. Then, the macro train schedule will be optimized again and some other conflicts may be resolved with the updated constraints. If some conflicts still exist, the conflicting train pair with the earliest occurrence time will be selected again until all conflicts are resolved.

Here, we introduce $y(I, k_m, k_n)$ as the decision variable of scheduling priority for a conflicting train pair (k_m, k_n) at station node I. In the initialization step, we set $y(I, k_m, k_n) = -1$ for all branching variables, which means that all trains will choose routes following FCFS strategy. If train k_m occupies a conflicting turn-out resource in priority than k_n, then $y(I, k_m, k_n) = 1$; otherwise, $y(I, k_m, k_n) = 0$. Thus, left and right child nodes of the search tree can be generated based on the value of $y(I, k_m, k_n)$ (=1 or 0). Under this branching scheme, constraint updating will be made according to the discussion in Section 5.3.

5.2.2. Bounding rules

We use the following three rules as the bounding criterion to prune unnecessary branching nodes, i.e., (1) the scheduling process of all trains cannot be completed within the predefined time horizon, (2) the total cost with the feasible solution of the current branching node is greater than the best objective value among the encountered BB nodes, and (3) the Lagrangian LB with the macro train scheduling problem of the current branching node is greater than the global UB. When one of above criteria is met, the current branching node will be deleted.

5.3. Constraint updating

When the priority of a train pair (k_m, k_n) has been determined by the BB method, the macro train timetable should be rescheduled using the constraint updating procedure. In this section, we introduce three basic constraint updating methods to improve the feasibility of train schedules by extending constraints (9) and (10). To begin with, we consider the segments $(I-1, I^A)$ and $(I^D, I+1)$ connected to the conflict station I, where $I^D(I^A)$ is the departure(arrival) event of station I. We do not change the subscripts of departure and arrival events at stations $I-1$ and $I+1$ as they are not crucial stations in the following discussion. Then, Eqs. (9) and (10) can be rewritten to Eqs. (31)–(32).

$$\sum_{T' \in \mathcal{T}^M} x\left(k, I^D, I+1, T, T'\right) = d\left(k, I^D, I+1, \tau\right) \quad \forall \tau \in \left[T, T + h^D(I^D, I+1)\right), T \in \mathcal{T}^M, k \in K \tag{31}$$

$$\sum_{T \in \mathcal{T}^M} x\left(k, I-1, I^A, T, T'\right) = a\left(k, I-1, I^A, \tau'\right) \quad \forall \tau' \in \left[T', T' + h^A(I-1, I^A)\right), T' \in \mathcal{T}^M, k \in K \tag{32}$$

For clarity, we further rewrite Eq. (31) into the following form in order to highlight the index of focused train pair k_m and k_n.

$$\sum_{T' \in \mathcal{T}^M} x\left(k_m, I^D, I+1, T, T'\right) = d\left(k_m, I^D, I+1, \tau\right) \quad \forall \tau \in \left[T, T + h^D(I^D, I+1, k_m, k_n)\right), T \in \mathcal{T}^M \tag{33}$$

Similarity, constraint Eq. (32) on arrival events can be reformulated as

$$\sum_{T \in \mathcal{T}^M} x\left(k_m, I-1, I^A, T, T'\right) = a\left(k_m, I-1, I^A, \tau'\right) \quad \forall \tau' \in \left[T', T' + h^A(I-1, I^A, k_m, k_n)\right), T' \in \mathcal{T}^M. \tag{34}$$

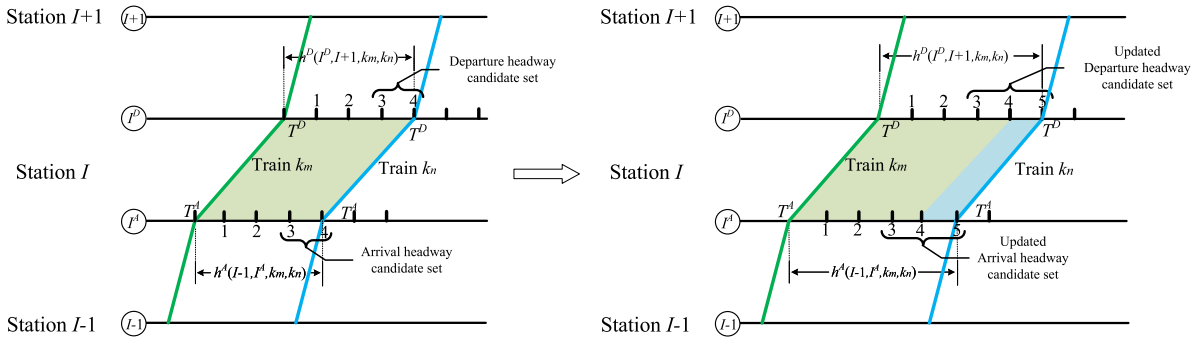


Fig. 15. Constraint updating for the conflicting train pair traveling in the same direction of case I.

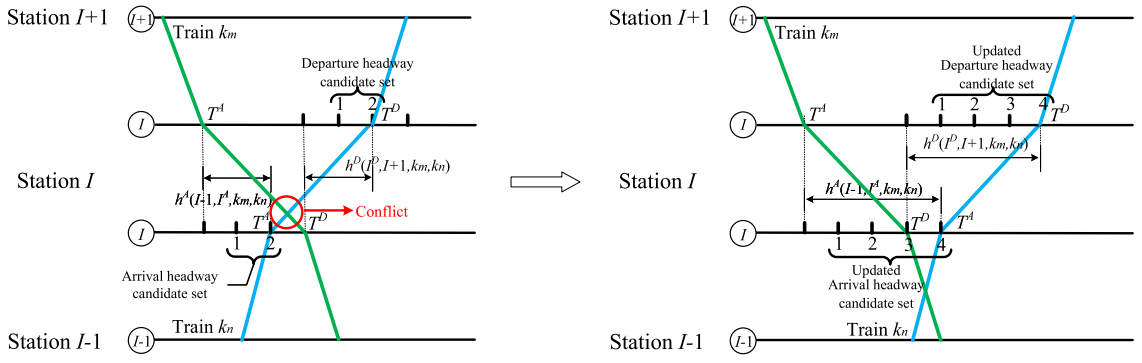


Fig. 16. Constraint updating for the conflicting train pair traveling in the opposite direction of case I.

In the following constraint updating procedure, we will update constraints by enlarging or reducing the candidate sets of arrival headway $h^A(I-1, I^A, k_m, k_n)$ and departure headway $h^D(I^D, I+1, k_m, k_n)$. For some complicated scenarios, the timetable planning time horizon should be extended. For example, in this illustrative section, each enlarging (or reducing) step is set to 1 min once, that is, one more element should be added to the candidate headway set for each time increment of 1 min, and the upper bound of enlarged headway is set to the dwell time of the adjacent previous train. Next, we aim to detail the constraint updating process for different cases.

Case I: Enlarging the candidate set of headway in tight-capacity stations. When trains k_m and k_n conflict at station I in the micro STN, we will update constraints (33) and (34) by enlarging the foregoing candidate sets. As illustrated in Fig. 15, for two trains traveling in the same direction, the initial headway candidate sets are $\{3,4\}$ for both $h^A(I-1, I^A, k_m, k_n)$ and $h^D(I^D, I+1, k_m, k_n)$. When the enlarging step size is set to 1 min, the both headway candidate sets for these two conflicting trains will be updated to $\{3,4,5\}$. Fig. 16 illustrates the conflict between two trains traveling in the opposite directions. Due to some station operation requirements, only one of the two trains can occupy the resources of station I at one time (especially at its junction). Under such a circumstance, a feasible micro train route plan can be obtained by updating the candidate sets of headways using the predetermined priority along with the branch-and-bound algorithm. In Fig. 16, the headway candidate sets corresponding the opposite train pair (k_m, k_n) are updated from $\{1,2\}$ to $\{1,2,3,4\}$ for $h^A(I-1, I^A, k_m, k_n)$ and $h^D(I^D, I+1, k_m, k_n)$, respectively.

Case II: Reducing the candidate set of headway in rich-capacity stations. Typically, railway stations with updated constraints will have more available space–time resources to schedule trains. In this case, it is easy to obtain a feasible conflict-free solution. Nevertheless, a high resource capacity redundancy might result in the wasted capacity. To avoid this, for the actual scheduled headway that does not reach the maximum value in the candidate set, the candidate set of headway would be reduced in such rich-capacity stations. However, the minimum element of updated candidate sets of $h^A(I-1, I^A, k_m, k_n)$ and $h^D(I^D, I+1, k_m, k_n)$ should not be less than the minimum safety headway. Taking Fig. 17 for instance, the initial headway candidate sets of the two trains traveling in the same direction are $\{3,4\}$, i.e., $h^A(I-1, I^A, k_m, k_n)$, $h^D(I^D, I+1, k_m, k_n) \in \{3,4\}$. When the headway reduction step is set to 1 min, the headway candidate sets will be updated to $\{3\}$, where 3 min is the minimum headway.

Case III: Extending the planning time horizon. If neither of cases I and II can resolve the conflicts effectively, it indicates that the current resources cannot guarantee the requirement of the predefined stop pattern. Therefore, it is necessary to extend the planning time horizon gradually in order to match the line planning scheme. As illustrated in Fig. 18, no available headway candidate sets after enlarging or reducing can be found to make solutions feasible. As some headway set still needs to be enlarged

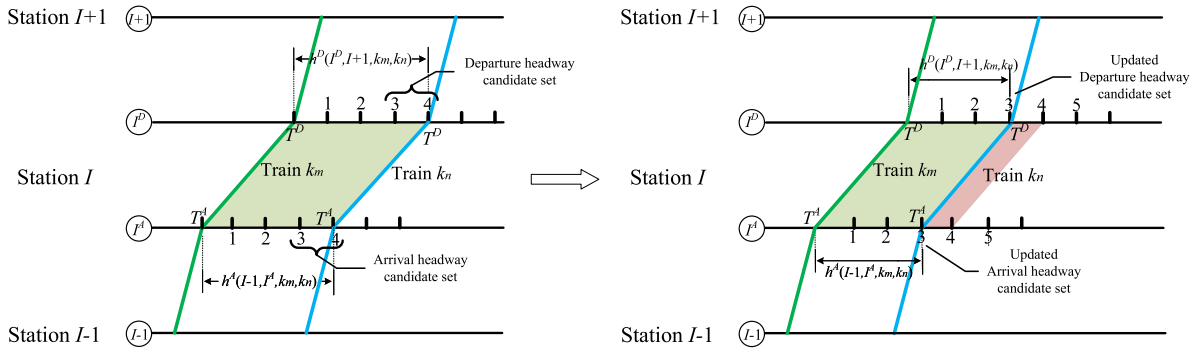


Fig. 17. Constraint updating for conflicting train pair traveling in the same direction of case II.

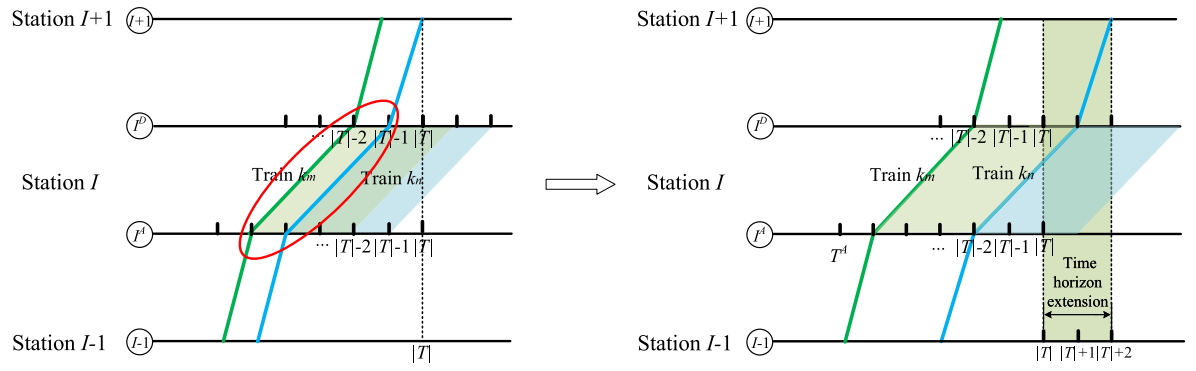


Fig. 18. Extending the planning time horizon of case III. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(such as the instance in the red circle of Fig. 18), the planning time horizon will be extended from $|T|$ to $|T|+2$ to ensure the last train has a complete feasible route in case that its schedule is out of the planning time horizon.

6. Numerical experiments

In this section, we aim to test the effectiveness and performance of the proposed model and algorithm with a series of numerical experiments using two different railway networks. For the small-scale network, the microscopic model will also be solved by Gurobi 9.5.1 directly for comparison. As the real-world case cannot be solved efficiency by Gurobi solver, we proceed the performance analyses by using the proposed algorithms only. All numerical experiments are implemented in Python 3.8.5 on a computer with Intel(R) Core i5-10300H@2.50 GHz CPU and 16 GB RAM.

6.1. Small-scale case study

In this set of experiments, we consider a small railway corridor with three stations and two segments, as shown in Fig. 19. In the operations, the departure–arrival tracks inside stations W6, W7 and W8 can provide services for trains traveling in the same direction by default (i.e., the fixed usage mode of tracks), while those tracks at stations W7 can be occupied by trains traveling in the opposite directions (i.e., the flexible usage mode of tracks) under flexible usage scenarios. To balance the computation efficiency and accuracy with respect to this illustrative case, the planed time resolutions are set as 1 min in the macro STN and 1s in the micro STN, respectively. Table 5 lists some required parameters with respect to the railway networks and train operations.

6.1.1. Computational results

With the experiment parameters mentioned above, we consider an all-stop line planning scheme shown in Fig. 20, which involves five downstream trains (i.e., Trains 0,1,2,4,5) and one upstream train (i.e., Train 3). To illustrate the detailed branch-and-bound solution process conveniently, the dwell time region (i.e., 9~11 min) is set to be greater than the minimum headway (i.e., 3 min), corresponding to a stressful capacity environment for the scheduling problem. Besides, the Lagrangian multiplier of each arc is set as 2 in the initial step (Zhang et al., 2019c).

In this set of experiments, only departure–arrival tracks at station $W7$ can provide services for trains traveling in the opposite directions when the flexible usage mode of tracks is considered. Thus, the branch-and-bound process can only be conducted at

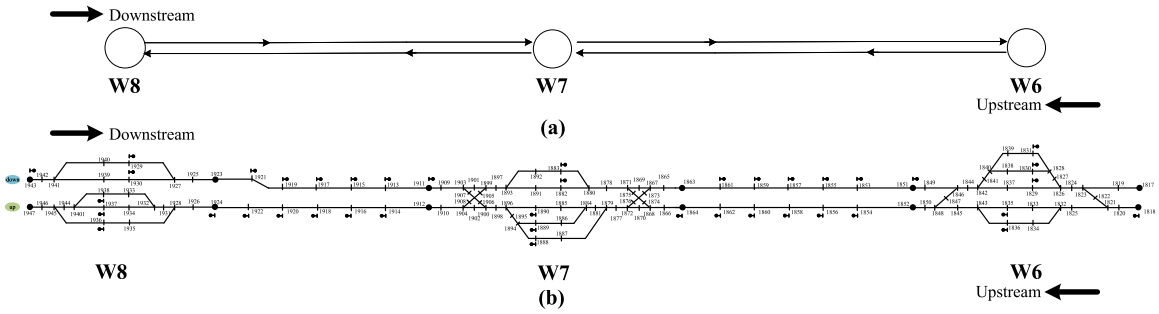


Fig. 19. A small-scale physical railway network in macro (a) and micro (b) levels.

Table 5
Parameters of railway networks and train operations.

Parameter	Value
Minimum departure headway $h^D(I,J)$	3 min
Minimum arrival headway $h^A(I,J)$	3 min
Station dwell time range	[9 min, 11 min]
Running time range on segment W8–W7	[8 min, 12 min]
Running time range on segment W7–W6	[16 min, 20 min]

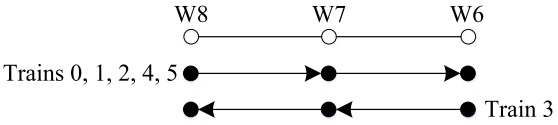


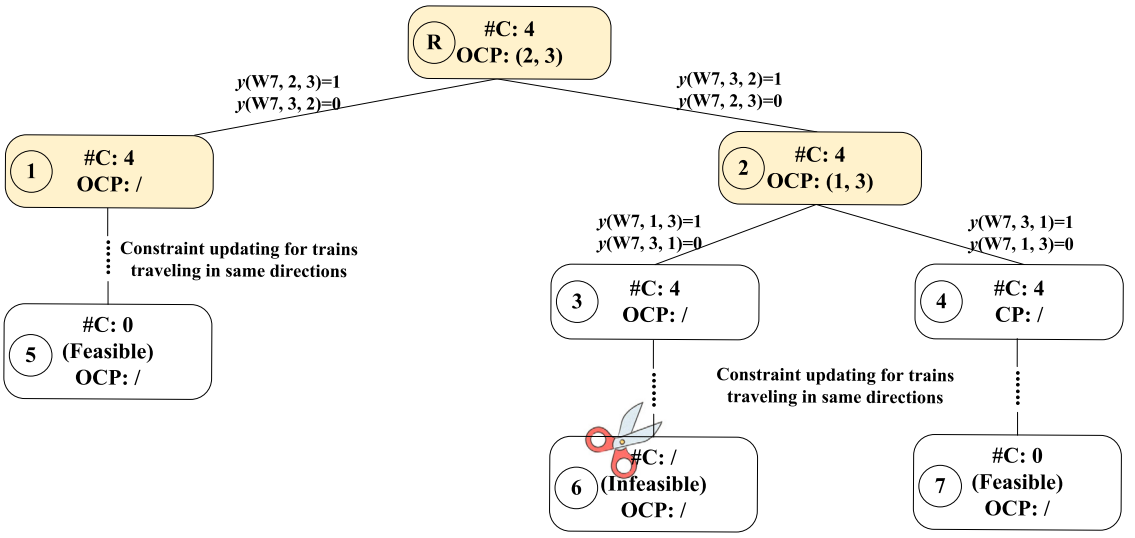
Fig. 20. Line planning scheme in the experiments.

station $W7$, which is illustrated in Fig. 21. The detailed solution information is attached to each node, including the total number of conflicts and the involved conflicting train pair for branching. Take the root node (R) for example, there are a total of four conflicts at station $W7$ (i.e., #C=4). One of the conflicts involves a train pair (train 2, train 3), which is shown by using the dotted line in Fig. 22. When multiple conflicting train pairs are detected, the branching rule is applied to the first-detected conflicting train pair by default. We make two branches from this node since only branching process is needed for the opposite conflicting train pair. Specifically, in branch 1, train 2 traverses the conflicting point before train 3 (see Fig. 22(a)), i.e., y(W7,2,3)=1 and y(W7,3,2)=0, and in branch 2, train 3 traverses the conflicting point before train 2 (see Fig. 22(b)), i.e., y(W7,3,2)=1 and y(W7,2,3)=0. After branching, the priority between trains 2 and 3 traveling in the opposite directions (i.e., variable $y(I,k,k')$) can then be specified.

Note that, the conflicting train pairs might also exist in the same direction. As shown in Fig. 21, after opposite train pairs (2,3) and (1,3) are fixed in sequence, there are still some conflicts for trains traveling in the same direction due to the inadequate headway. After completing the constraint updating procedure for the conflicting train pairs traveling in the same direction, the number of conflicts in this node decreases to 0, and then a feasible solution is obtained with no conflict. Specifically, four conflicts are observed in BB node 4 (#C: 4), including 0 conflicting train pair in the opposite direction (OCP: /) and 4 conflicting train pairs in the same direction (#C-OCP=4). After processing the constraints updating procedure for this conflicting train pair from node 4 to node 7, all conflicting train pairs have been resolved (#C=0, OCP=0) to generate a feasible solution. As shown in Fig. 23, the train trajectories with dot lines are adjusted to full lines. Additionally, node 6 is pruned in the bounding process due to the infeasible macroscopic train schedule solution.

6.1.2. Performance analyses over different scenarios

In general, a line planning scheme with complex operation patterns will potentially cause the difficulties in the scheduling and routing process. Thus, it is necessary to investigate effectiveness of the proposed model and algorithm against a variety of diverse line planning schemes. Two line planning schemes and corresponding parameters of railway networks and train operations are considered in Fig. 24 and Table 6. In order to improve the solving efficiency of the proposed algorithm and test the robustness with different macroscopic and microscopic resolution combinations, the planed time resolutions are set as 1 min in the macro STN and 3s in the micro STN, respectively. Table 7 lists the computation results under different tested scenarios with respect to various line planning schemes, platform designs, and station operation modes. Moreover, column “LPS” denotes the line planning scheme; column “SOM” denotes the station operation mode in W7, indicating whether trains are allowed to use opposite departure–arrival tracks (i.e., flexible usage mode) or not (i.e., fixed usage mode). In addition, the optimal objective values, optimal gaps, and comparisons with the results by Gurobi solver are also included, where column “Gap 1” denotes the relative gaps in the proposed algorithm.



Notes:
®, ①, ②: No. of branching tree node.
#C: Numbers of total conflicts.
OCP: Opposite conflicting train pairs at station W7

Fig. 21. Branch-and-bound searching process.

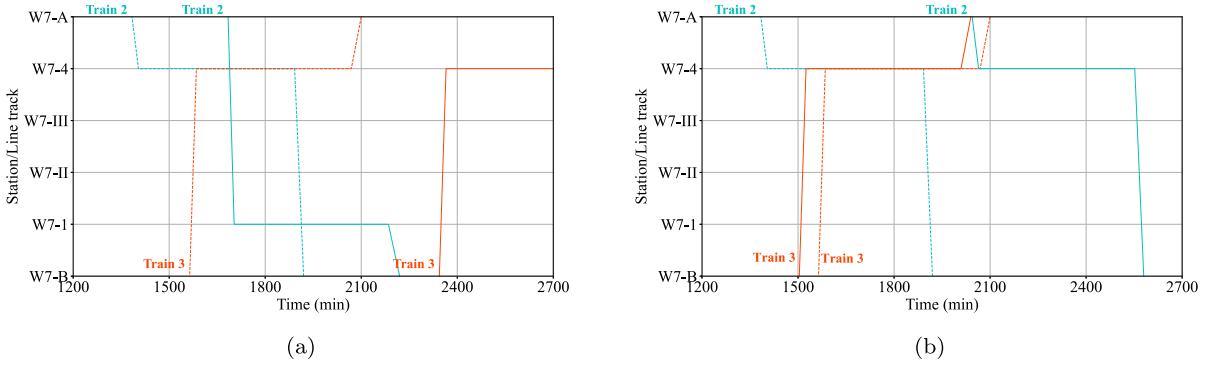


Fig. 22. Branching results for train conflict pair (train 2, train 3) in the root node.

Table 6
Parameters of railway networks and train operations.

Parameter	Value
Minimum departure headway $h^D(I, J)$	3 min
Minimum arrival headway $h^A(I, J)$	3 min
Station dwell time range	[5 min, 6 min]
Running time range on segment W8–W7	[8 min, 10 min]
Running time range on segment W7–W6	[8 min, 10 min]

In addition, to provide a benchmark for the computation results, we also consider the micro train routing model in the micro STN. We use Gurobi solver to solve this simplified model directly for computational efficiency. The computing results are displayed in columns "Gurobi solver" in Table 7. It is worth mentioning that the train timetable at the macroscopic level can be deduced from the generated train routes at the microscopic level. In other words, the train routing problem can be viewed as a relaxed model without considering macro constraints, which can obtain better objective value than the coupled "macro–micro" model. Typically, this solution process is effective when the train routing model is moderate in scale.

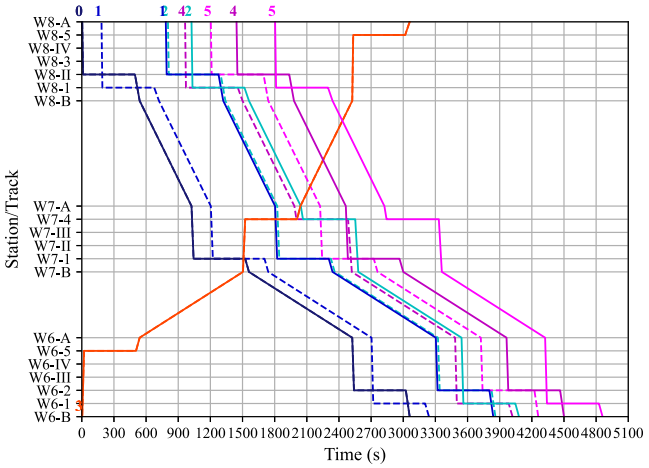


Fig. 23. Result for the CU of the conflicting train pair traveling in the same direction.

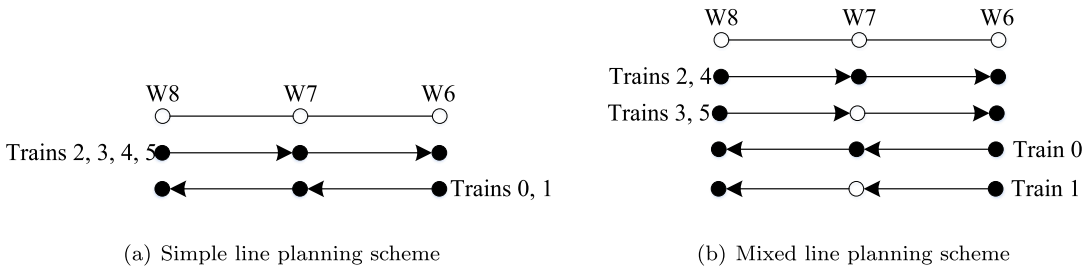


Fig. 24. Line planning schemes.

Next, we will analyze the computational performance of the proposed algorithm and Gurobi solver with a total of 16 experiment instances. In these experiments, it is clear that the proposed method can find (near) optimal solutions using much less time than Gurobi solver for all the cases (see Table 7 for detail). As mentioned above, the optimal objectives of the micro routing model are found to be better than those of the coupled "macro–micro" model. The difference between these two models essentially result from the inconsistent network resolution. In detail, since the time granularity of the micro STN is set as 3 s, when solving the micro train routing model by Gurobi solver, the arrival or departure times of a train at the boundary of stations are probably not consistent with the time granularity in the macro STN (i.e., 60 s). In this case, extra waiting times for this train will occur if we aim to deduce the corresponding train timetable based on the micro routes, which results in the different objective values between the proposed algorithm and Gurobi solver. Therefore, the optimal objective values obtained by the proposed algorithm and Gurobi might be not comparable due to the different time resolutions. However, the gap between these two optimal objectives is still meaningful for evaluating the solution quality of the coupled macro–micro model (see Gap 3 in Table 7).

Some key operation results are also summarized in Table 9 in Appendix, including track allocations in station W7, departure times at beginning stations, and arrival times at ending stations for different trains. For example in the scenario instance "S_1U1D_Fixed", there are 1 track in the upstream direction and 1 track in the downstream direction under the fixed usage for departure–arrival tracks at station W7. We notice that some experiments return the same optimal objective value due to the form of the objective function. Specifically, the total link traveling time is regarded as the objective function in the formulated mathematical model. Therefore, the computing results tend to increase the headway between trains so as to avoid the increase of waiting or traveling costs. For some solutions with the same objective, although the corresponding macroscopic train trajectories are similar, the track choices of trains at the micro level are actually different. Note that the objective function does not specifically consider such factors like track selection, when the line planning scheme is the same, the optimal objective value obtained might also be the same.

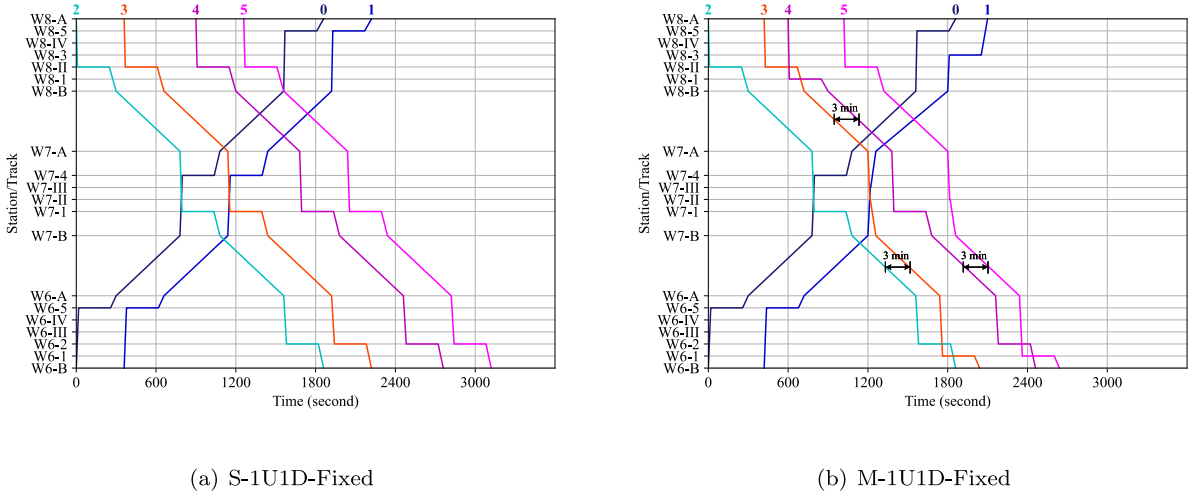
Additionally, it is worth noting that the complexity and scale of the problem will increase dramatically when departure–arrival tracks could be used by trains traveling in the opposite directions. For instance, the computing time of case "S-1U1D-Flexible" rises more than 10 times than that of case "S-1U1D-Fixed" by Gurobi solver, as shown in Table 7. Next, we will detail the performance analyses with respect to different model setups, including line planning schemes, railway station layouts, and railway station operation modes.

(1) Performance analyses with respect to line planning schemes.

Table 7
Computing results for joint optimization in different scenarios.

Scenario No.	Scenario feature	LPS	# of track (W7)	SOM	# of constraint updating			MSTNA			Gurobi solver			Gap 3
					W8	W7	W6	Obj. val.	Running time	Gap 1	Obj. val.	Running time	Gap 2	
1	S_1U1D_Fixed	Simple	1U/1D	Fixed	0	2	2	11160	18.51 s	1.85%	10656	376.87 s	0.0%	4.52%
2	S_1U1D_Flexible	Simple	1U/1D	Flexible	0	2	2	11160	33.09 s	2.18%	10620	4965.44 s	0.0%	4.84%
3	S_1U2D_Fixed	Simple	1U/2D	Fixed	0	2	2	11160	31.39 s	2.72%	10656	274.22 s	0.0%	4.52%
4	S_1U2D_Flexible	Simple	1U/2D	Flexible	0	2	2	11160	31.91 s	2.72%	10632	6885.51 s	0.0%	4.73%
5	S_2U1D_Fixed	Simple	2U/1D	Fixed	0	5	2	11160	36.49 s	4.14%	10656	372.60 s	0.0%	4.52%
6	S_2U1D_Flexible	Simple	2U/1D	Flexible	0	2	2	11160	22.43 s	2.18%	10632	3085.70 s	0.0%	4.73%
7	S_2U2D_Fixed	Simple	2U/2D	Fixed	0	0	2	11160	21.92 s	3.30%	10656	307.73 s	0.0%	4.52%
8	S_2U2D_Flexible	Simple	2U/2D	Flexible	0	0	2	11160	22.24 s	3.40%	10632	3011.59 s	0.0%	4.73%
9	M_1U1D_Fixed	Mixed	1U/1D	Fixed	0	0	0	10500	7.11 s	1.30%	9999	461.82 s	0.0%	4.77%
10	M_1U1D_Flexible	Mixed	1U/1D	Flexible	0	0	0	10500	8.16 s	1.19%	10284	5147.31 s	0.0%	2.06%
11	M_1U2D_Fixed	Mixed	1U/2D	Fixed	0	0	0	10500	5.77 s	1.19%	10083	487.67 s	0.0%	3.97%
12	M_1U2D_Flexible	Mixed	1U/2D	Flexible	0	0	0	10500	5.76 s	1.19%	(UB)10146 (LB)9123	10800 s	10.08%	3.37%
13	M_2U1D_Fixed	Mixed	2U/1D	Fixed	0	0	0	10500	5.62 s	1.19%	10023	486.55 s	0.0%	4.54%
14	M_2U1D_Flexible	Mixed	2U/1D	Flexible	0	0	0	10500	5.77 s	1.19%	(UB)10146 (LB)9123	10800 s	10.08%	3.37%
15	M_2U2D_Fixed	Mixed	2U/2D	Fixed	0	0	0	10500	5.90 s	1.19%	10080	460.84 s	0.0%	4%
16	M_2U2D_Flexible	Mixed	2U/2D	Flexible	0	0	0	10500	6.47 s	1.19%	(UB)10146 (LB)9123	10800 s	10.08%	3.37%

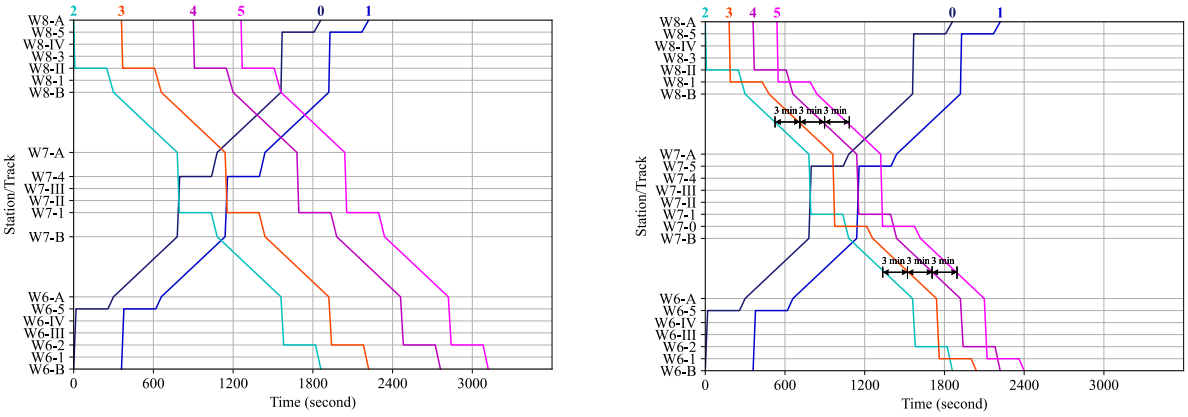
Notes:
LPS: Line planning scheme (Simple: all-stop mode (Fig. 24(a)); Mixed: all-stop mode and stop-skipping mode (Fig. 24(b))).
of tracks (W7): Number of departure–arrival tracks for trains traveling in the upstream direction (U) or in the downstream direction (D) at station W7.
SOM: Station operation mode (Fixed or flexible usage for departure–arrival tracks).
of constraint updating: Number of constraint updating iterations.
Obj. val.: Returned best objective value.
S_1U1D_Fixed: Simple line planning scheme, i.e., 1 track in the upstream direction and 1 track in the downstream direction at station W7 under the fixed usage for departure–arrival tracks.
Gap 1: Relative gap between UB and LB in the proposed algorithm (Coupled "macro–micro" model).
Gap 2: Relative gap between UB and LB in Gurobi (Micro model).
Gap 3: Relative gap between the Obj. vals between Gurobi and MSTNA.



(a) S-1U1D-Fixed (b) M-1U1D-Fixed

Fig. 25. Comparison of joint optimization results under different line planning schemes.

The complexity of the proposed model is closely related to the line planning schemes on the railway corridor. If involved trains are operated with the mixed line planning scheme in Fig. 24(b), the throughput capacity might be improved within the given time horizon. In comparison, the simple line planning scheme in Fig. 24(a) with all-stop modes might lead to the loss of throughput capacity to some extent. Besides, for instance in Fig. 25(b), when trains with the stop-skipping and all-stop modes are scheduled alternatively, the joint optimization solution tends to fully utilize the throughput capacity of this railway corridor in comparison with the case in Fig. 25(a). In Fig. 25(a), only one departure–arrival track is available for individual directions at station W7, which is the main bottleneck to the entire railway corridor. Thus, all trains should be scheduled to stop and pass through station W7 one by one, which consequently leads to the increase in headway and reduction in capacity for other two stations. In Fig. 25(b), when the line planning schemes for some trains are adjusted to the operation mode of skipping station W7, the headway of train pair (3, 4) on segment W8–W7 and train pair (2, 3) on segment W7–W6 can reach the minimum headway requirement (i.e., 3 min), which can fully use the throughput capacity of the corridor. Then, the scheduling duration for scheduling is reduced from 3120s under the simple line planning scheme to 2640s under the mixed line planning scheme. Therefore, for stations with limited resources, the railway operators should pay more attention to homogeneous factors in the line planning schemes (e.g., the train stop plan, the number of fast/slow trains) so as to utilize the railway throughput capacity sufficiently.



(a) S-1U1D-Fixed

(b) S-2U2D-Fixed

Fig. 26. Comparison of joint optimization results between different station layouts.

(2) Performance analyses with respect to railway station layout. The station layout has a crucial impact on the micro train route selection and constraint updating process. The capacity of an entire railway corridor usually depends on the tightest station or junction capacity on this line. When the dwell time at some stations is longer than the segment headway, a railway station which has multiple departure–arrival tracks can improve the railway throughput capacity and reduce the problem difficulty. For instance, there is a significant difference in the throughput capacity between cases "S-1U1D-Fixed" in Fig. 26(a) and "S-2U2D-Fixed" in Fig. 26(b). Clearly, since no additional tracks can be chosen in station W7 in Fig. 26(a), all trains must dwell at this station one by one, which does not fully utilize the capacities of stations W6 and W8. In comparison, as multiple dwell tracks are available at station W7 in Fig. 26(b), all downstream trains can run with the minimum headway (i.e., 3 min). Nevertheless, for case "S-2U2D-Fixed", station W7 does not fully utilize its capacity even with multiple departure–arrival tracks due to the bottleneck on the limited upstream departure tracks at station W6. Moreover, the case with two upstream tracks and two downstream tracks only needs to execute two iterations of the constraint updating process to obtain a near-optimal solution while the case with one upstream track and one downstream track needs four iterations, which takes more computational time. The above computational results show that in order to reduce the waste of redundant capacity for railway stations, railway station layouts should be coordinated as part of the planning process before construction.

(3) Performance analyses with respect to railway station operation modes. Under the flexible usage mode of departure–arrival tracks, the throughput capacity of the entire railway corridor can be improved to a great extent. In this case, some trains need to use the opposite departure–arrival tracks to ease the stress on the station capacity. Accordingly, the safe headway between trains traveling in the opposite directions and those on bi-direction tracks should be considered. For instance, with the joint optimization result of the case in Fig. 27(a), station W7 becomes a main bottleneck of the entire railway corridor, and trains 4 and 5 traveling in the downstream direction have to stop in sequence when the opposite tracks cannot be utilized. Moreover, the operation time horizon lasts for 37 min in the upstream direction and 52 min in the downstream direction, respectively. In Fig. 27(b), train 3 can arrive at the upstream track at station W7 and keep a smaller headway with train 4 on segments W8–W7 and W7–W6 due to the flexible usage of departure–arrival tracks. This can remarkably improve the throughput capacity, and then reduce the operation time horizon from 52 min to 50 min. Thus, a rational design of the station operation plan along with the flexible track usage could greatly improve the station capacity utilization rate, above all at some stations with complex topological structure.

6.2. Real-world case study based on the RAS dataset

In this section, the performance of the proposed model and algorithm is demonstrated by a set of large-scale experiments using a real-world dataset from the INFORMS RAS in 2016 (Meng, 2016; Lu and Zhou, 2022). Fig. 28 shows a real-world railway network with detailed segment lengths and station layouts, which is composed of 8 physical stations and 7 bidirectional segments. The planning time horizon is set to 300 min (i.e., 18,000 s) in both upstream and downstream directions. The departure–arrival tracks in stations $W1{\sim}W4$ can provide services for all trains traveling in upstream and downstream directions (i.e., Flexible $W1{\sim}W4$) while the departure–arrival tracks in other four stations are prohibited to be occupied by trains traveling in the opposite directions (i.e., Fixed $W5{\sim}W8$). In this experiment, a total of 48 trains are taken into consideration, including 24 trains traveling in the downstream direction and 24 trains traveling in the upstream direction, respectively. These trains are categorized into high-speed trains and low-speed trains, i.e., the mixed line planning scheme is adopted here. We consider five scenarios to test performances of the proposed approach under different numbers of high-speed and low-speed trains with the line planning scheme shown in Fig. 29. Specifically, denote N_1, N_2, N_3 and N_4 as the number of planned trains with different line planning schemes G_1, D_1, G_2 and D_2. In

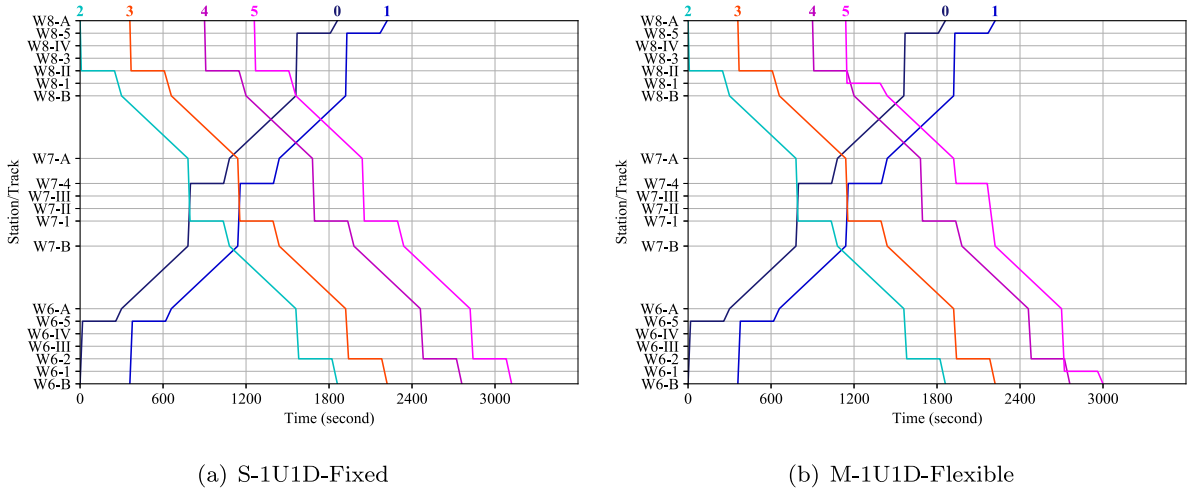


Fig. 27. Joint optimization result comparison between different station operation modes.

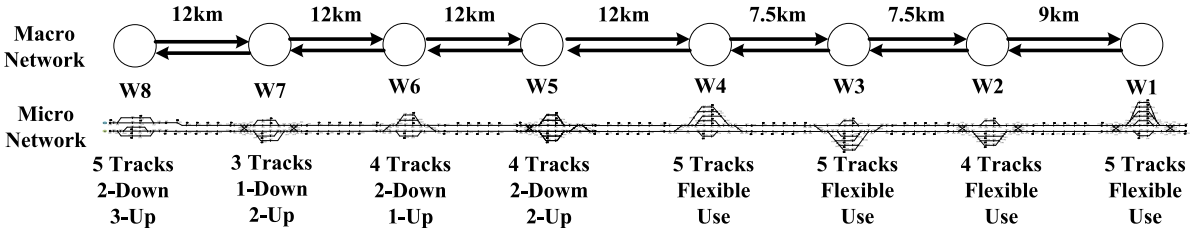


Fig. 28. An overview of the railway network.

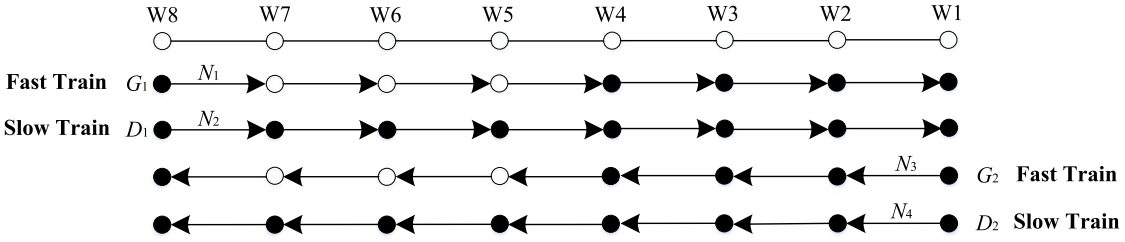


Fig. 29. The line planning schemes for the real-world case study.

order to improve the computational efficiency in the solution process, three acceleration techniques are embedded into the algorithm: (1) solving micro train routing problems inside each station parallelly; (2) solving schedules of individual trains parallelly in LR lower bound generation processes; (3) skipping the priority rule algorithm in the early iterations of implementing the LR algorithm in solving the macro train scheduling problem; (4) applying the diving strategy for BB nodes with the least objective value in the branch-and-bound tree.

Table 8 summarized the detailed network information and computation results from this real-world case study. For these five scenarios, the UB jumps with the number of slow trains, as more slow trains will lead to more microscopic conflicts at early iterations. The near-optimal solution can be found within 40 min (2400s) with the acceleration techniques. Moreover, scenario 4 is used to make additional evaluations so as to obtain insights for railway management. Specifically, Fig. 30 visualizes the computational result with respect to scheduling and routing plans, with different colors indicating individual space–time trajectories of different trains. In this case, the trains traveling in the upstream direction need to be scheduled in time horizon [0 min, 290 min] (i.e, the duration is 17,400s) while the trains traveling in the downstream direction complete operations in time horizon [0 min, 260 min] (i.e., the duration is 15,600 s). When the same number of trains are dispatched in two directions, the throughput capacity in the downstream direction is much larger than that in the upstream direction.

Next, we will detail the performance analyses through highlighting some local characteristics of the joint optimization solution. As a result of stop-skipping operation modes at stations $W1 \sim W4$, some trains traverse on many segments following the minimum

Table 8
Summary of the joint optimization for the real-world case study.

Scenario No.	Line planning scheme				# of CU	Lower bound (LB)	Upper bound (UB)	Gap (%)	CPU time (s)
	Upstream		Downstream						
	Fast (N_1)	Slow (N_2)	Fast (N_3)	Slow (N_4)					
1	6	18	6	18	7	282793.70	288000	1.84	1022.55
2	9	15	9	15	18	277733.95	283680	2.10	2223.03
3	12	12	12	12	6	275363.47	279360	1.43	1316.45
4	15	9	15	9	9	270695.55	275040	1.5	2024.31
5	18	6	18	6	6	266649.80	270720	1.58	1274.19
							Ave.	1.69	1572.1

Notes:
N_i: Number of trains with respect to line planning scheme i.
of CU: Number of constraint updating iterations.
Lower bound (LB): Lagrangian lower bound for the macro train scheduling model at the optimal BB node.
Upper bound (UB): Optimal objective value for the coupled macro–micro model.
Gap: $gap = \frac{UB-LB}{UB} \times 100\%$.

headway, lead to highly parallel spatial–temporal trajectories. The differences in throughput capacity under different line planning schemes (e.g., cases A and B as shown by the blue and yellow regions in Fig. 30) can be easily identified. As there is only one downstream departure–arrival track ($W7-1$) at station W7, this station might become a local bottleneck of the railway line. In case A, trains 2 and 3 are required to stop at station W7 in the line planning scheme. Thus, the headway between trains 2 and 3 is enlarged to 6 min to resolve the conflict at station W7, which leads to the wasted capacity on segment $W8$–$W7$. On the other hand, only train 6 is required to stop at station W7 in case B. Then, trains 4∼7 are allowed to follow the minimum headway on segment $W8$–$W7$, which can fully utilize the throughput capacity.

In general, the overall throughput capacity might be bounded by a station with the tightest capacity along a rail line. Case C (i.e., the red region in Fig. 30) shows that there are two upstream tracks available at stations W5 and W7 but only one at station W6. Therefore, trains 42 and 43 cannot follow the minimum headway to avoid the emergence of conflicts, i.e., stations W5 and W7 cannot fully utilize their capacities although they have multiple departure–arrival tracks. On the contrary, the flexible usage mode of departure–arrival tracks plays a vital role in improving the throughput capacity. In case D (i.e., the purple region), train 26 can occupy track W4-4 efficiently when the occupation block of train 1 traveling in the opposite direction is cleared. Besides, all departure–arrival tracks (i.e., W4-1, 2, 3, 4, 7) are utilized flexibly by trains traveling in different directions during a busy time period in case E (i.e., the green region), which greatly improves the operation efficiency.

Fig. 31 displays the convergence of the proposed algorithm with respect to the number of conflicts over different iterations, where an iteration refers to the one-time execution of the branch-and-bound and constraint updating procedures. Take the blue line (i.e., station W2) as an example, a total of six conflicts (including conflicts in the same and opposite directions) occur in the first iteration. When the branch-and-bound and constraint updating procedures are applied to the basic solution in the root node, the number of conflicts roughly follows a downward trend as the number of iterations grows. After 9 iterations, all coupling constraints in the multi-resolution space–time network are satisfied and a near-optimal joint optimization solution is obtained. Moreover, the BB and CU procedures are mainly carried out for station W2, indicating that this station is actually a critical bottleneck on the entire railway corridor. Thus, the extension at station W2 should be considered in priority in the future so as to improve the overall throughput capacity.

7. Conclusions and future research

In this paper, aiming to obtain a conflict-free train scheduling and routing scheme in both macro and micro railway networks, we tackled the joint optimization of the train scheduling and routing problem by using a coupled multi-resolution space–time network modeling framework. In the proposed model, a series of necessary safety and flow conversation constraints were considered in different resolution networks. Then, the coupling constraints were formulated between the macro and micro space–time arc selection decision variables. To solve the proposed model, an iterative branch-and-bound algorithm with the constraint updating approach was developed. The performance of the proposed model and algorithm was tested against several small cases and a real-world case with the RAS extended dataset.

It is worth mentioning that in comparison to the existing "macro–micro" approaches in the literature, our approach can handle this joint optimization problem in a multi-resolution coupled railway network without the requirement of a general MILP solver, thanks to the modeling framework and algorithm in the space–time network. We can deduce the following managerial insights from the perspective of operations. That is, the utilization of the line capacity can be improved by considering (a) the homogeneous factors in the line planning scheme: highly homogeneous line planning schemes might result in the parallel train schedules with high throughput capacities and reduce capacity losses of trains potentially caused by diverse stop patterns; (b) the coordination of station layout on the entire railway corridor: the throughput capacity of the entire railway line is usually determined by the station with the least capacity, so it is necessary to improve the capacity (e.g., increasing the number of tracks) of some bottleneck stations;

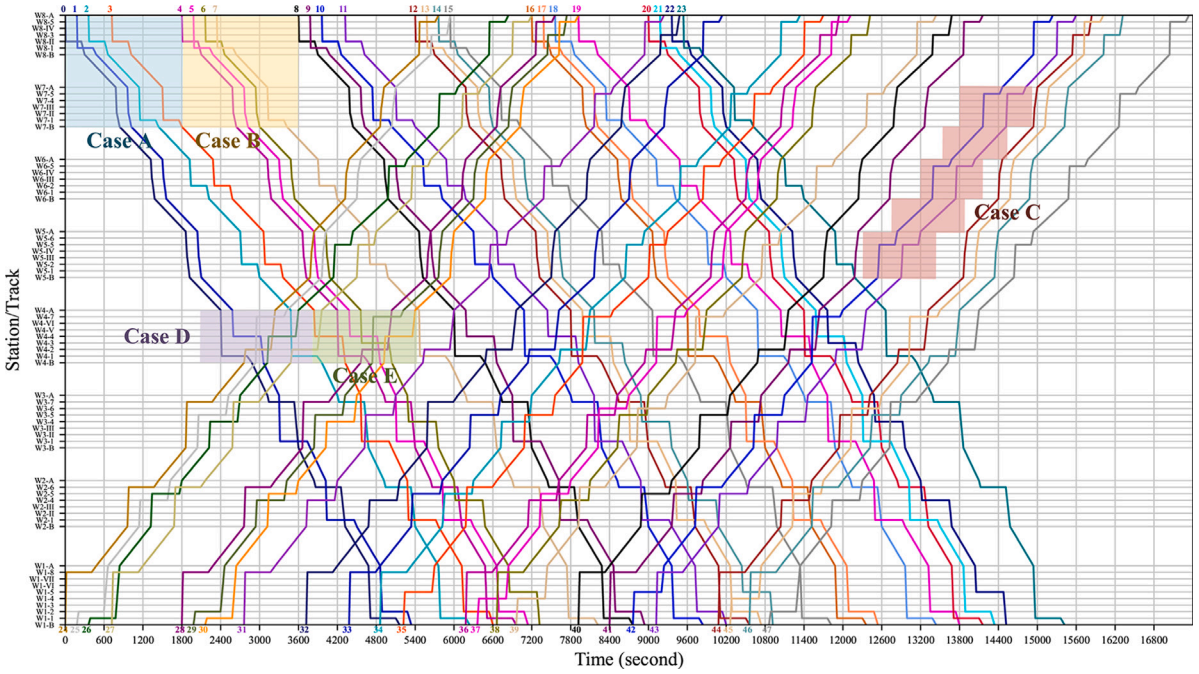


Fig. 30. Visualization for joint optimization result in the real-world case study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

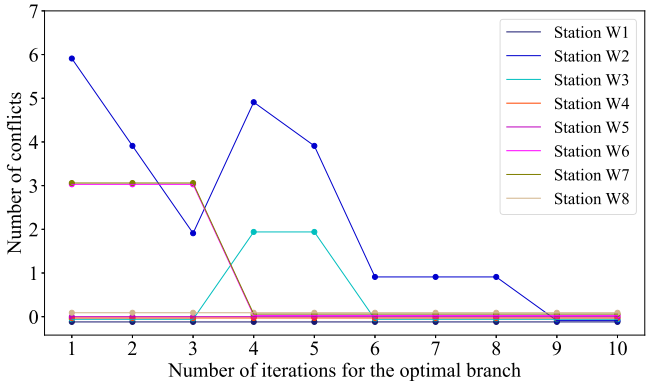


Fig. 31. Number of conflicts at each station with respect to different iterations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and (c) the rational design of the station operation plan with the flexible track usage: for an uneven schedule in upstream and downstream directions, operations with the flexible track usage mode can significantly improve the operation efficiency, especially in the microscopic railway network of bottleneck stations.

The limitations of our approaches are two aspects. (1) The computation for the real-world case study is still time-consuming because of the large-scale arc selection variables and possible BB nodes. (2) A much tighter LB for the coupled "macro–micro" model (Model M1) should be calculated, and more efforts can be made to further improve the solution quality for large cases. In addition, our further research might focus on the following topics. (1) Consider more practical railway operation modes in the joint optimization problem towards real-world cases, such as picking up and delivering train cars or scheduling locomotives. (2) Develop an efficient exact decomposition algorithm with more acceleration techniques to speed up the searching process and improve the solution accuracy. (3) Design some tight bounding rules to truncate unnecessary branches to enhance the solution quality through conflicting train pair enumeration.

Table 9
Summary for track allocations, departure/arrival times in different scenarios.

Scenario No.	Scenario feature	Track allocation						Train departure time (s)						Train arrival time (s)					
								From station W8			From station W6			From station W8			From station W6		
	Train ID	0	1	2	3	4	5	0	1	2	3	4	5	0	1	2	3	4	5
1	H-1U1D-Fixed	1	1	1	1	4	4	0	360	0	360	900	1260	1860	2220	1860	2220	2760	3120
2	H-1U1D-Flexible	1	1	1	4	4	4	0	360	0	360	900	1140	1860	2220	1860	2220	2760	3000
3	H-1U2D-Fixed	1	0	1	0	4	4	0	360	0	180	780	960	1860	2220	1860	2040	2640	2820
4	H-1U2D-Flexible	1	0	1	4	4	4	0	360	0	180	780	960	1860	2220	1860	2040	2640	2820
5	H-2U1D-Fixed	1	1	1	1	5	5	0	360	0	360	720	1080	1860	2220	1860	2220	2580	2940
6	H-2U1D-Flexible	4	4	1	4	5	5	0	360	0	360	540	780	1860	2220	1860	2220	2400	2640
7	H-2U2D-Fixed	1	0	1	0	5	5	0	360	0	180	360	540	1860	2220	1860	2040	2220	2400
8	H-2U2D-Flexible	4	1	4	1	5	5	0	360	0	180	360	540	1860	2220	1860	2040	2220	2400
9	NH-1U1D-Fixed	II	II	1	1	III	4	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
10	NH-1U1D-Flexible	II	II	1	4	III	4	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
11	NH-1U2D-Fixed	II	II	1	1	III	4	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
12	NH-1U2D-Flexible	II	II	1	4	III	4	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
13	NH-2U1D-Fixed	II	II	1	1	III	5	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
14	NH-2U1D-Flexible	II	II	4	5	III	5	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
15	NH-2U2D-Fixed	II	II	1	1	III	5	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400
16	NH-2U2D-Flexible	II	II	4	4	III	5	0	60	0	420	600	1020	1860	1500	1860	1800	2460	2400

Acknowledgments

This work was supported the National Natural Science Foundation of China (Nos. 71825004, 72288101, 72001019), and the State Key Laboratory of Rail Traffic Control and Safety, China (No. RCS2022ZZ001), Beijing Jiaotong University, China. The authors thank three reviewers for their helpful and valuable advices, and thank Prof. Xuesong Zhou from Arizona State University for helpful comments, which together improved this paper greatly.

Appendix. Summary for joint optimization results in different scenarios

See Table 9.

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